

10

Resolution Enhancement Technologies

In general, the optimum image quality for an arbitrary object is obtained by using a classical imaging system with zero aberrations. Semiconductor lithography, however, uses a very limited set of objects (arrays of lines and spaces, isolated lines, small square holes called contacts, etc.) in limited arrangements and orientations. In many cases, the sizes of these objects are also limited to set values. Thus, lithography can pose a unique imaging problem – given a single object (say, an array of contact holes of a given size), is it possible to design a special imaging system which is better than a classical imaging system for the printing of this one object? The answer is yes.

The design of an imaging system optimized for a reduced class of objects has been called *wavefront engineering* by one of the earlier pioneers in this area, Marc Levenson.¹ Wavefront engineering can be thought of in a number of different ways, but essentially it refers to manipulating the optical wavefront exiting the projection lens to produce improved images for certain types of objects. There are three basic ways of modifying the wavefront – manipulating the object (the mask), adjusting the illumination of the object, or modifying the wavefront directly with a pupil filter. The use of a pupil filter is difficult with current lens designs since the pupil is not directly accessible within the lens. Modified illumination and masks, however, are widely used in semiconductor lithography to improve imaging.

In its simplest form, an imaging system uses an object which is identical to the desired image. However, knowing the limitations of the imaging system, one can design a new object which produces an image more like the desired pattern. By adjusting the transmittance of the mask, one can improve the exposure latitude and depth of focus of a given image. Ideally, the spatial variation of both the phase and the transmittance of the mask would be modified in whatever way needed to improve the image. From a practical point of view though, masks for semiconductor lithography are essentially devices with discrete levels of transmittance (the simplest mask is binary, either 100 or 0% transmittance).

Each added level of transmittance adds considerably to the cost and complexity of the mask. For this reason, mask modification falls into two categories – mask shaping (also called *optical proximity correction*, or OPC) and *phase-shifting masks* (PSMs). Mask shaping adjusts the shape of the mask features to improve the shape of the resulting image (for example, by applying serifs to the corners of a rectangular pattern). The difficulty comes in predicting the needed mask shapes (a typical chip will require masks with hundreds of millions or billions of patterns that must be shaped). PSMs change (or possibly add to) the levels of transmittance on the mask to include phase differences between transmittance levels. Light shifted by 180° will interfere with unshifted light to produce well-controlled dark areas on the image. The difficulty again includes designing the mask, but also involves more complicated mask fabrication.

Modifying the illumination of the mask will also result in different wavefronts and thus different images. As we saw in Chapters 2 and 3, tilting the illumination can double the resolution of grating (line/space) patterns and can also improve the depth of focus of gratings of certain periods. This tilting, however, does not improve the performance of nonrepeating patterns. Thus, the direction (or directions) of the illumination can be customized for given mask features, but there is no one illumination direction which is best for all mask features. Modified illumination (also called *off-axis illumination* or OAI) can be combined with mask modification (OPC and possibly PSM) to produce results better than either approach alone, but with increasingly complicated processes.

Collectively, these three techniques, OPC, PSM and OAI, are known as *resolution enhancement technologies* (RETs). To see how they actually improve resolution, it is important to first carefully define what is meant by the term resolution.

10.1 Resolution

The resolution limit of optical lithography is not a simple function. In fact, resolution limits differ depending on the type of feature being printed. In general, however, there are two basic types of resolution: the smallest pitch that can be printed (the pitch resolution) and the smallest feature that can be printed (the feature resolution). While related, these two resolutions are limited differently by the physics of lithography, and have different implications in terms of final device performance. Pitch resolution, the smallest linewidth plus spacewidth pair that will print, determines how closely transistors can be packed together on one chip. This resolution has the greatest impact on cost per function and functions per chip. Feature size resolution determines the characteristics and performance of an individual transistor, and has the greatest impact on chip speed and power consumption. Obviously both are very important.

Pitch resolution is the classical resolution discussed in most optics textbooks and courses. It is governed by the wavelength of the light used to form the images and the numerical aperture of the imaging lens. The classical pitch resolution is given by a version of the Rayleigh resolution equation:

$$\text{pitch resolution} = k_{\text{pitch}} \frac{\lambda}{NA} \quad (10.1)$$

where λ is the vacuum wavelength of the lithographic imaging tool, NA is the numerical aperture, and k_{pitch} depends on the details of the imaging process. Ultimately, k_{pitch} can be as low as 0.5, but only with tremendous effort (as will be discussed below). Values of 0.7–1.0 are typical, with a trend toward decreasing values approaching the 0.5 limit.

While pitch resolution has a hard physical limit given by the Rayleigh equation, feature resolution is limited by the ability to control the critical dimension (CD) of the feature. As features are made smaller, control of the CD of that feature becomes harder. There is no hard cutoff, only a worsening of CD control as the feature size is reduced. Feature size control is governed by the magnitude of various process errors that inevitably occur in a manufacturing environment (such as focus and exposure errors), and the response of the process to those errors (see Chapter 8). In order to improve CD control, one must simultaneously reduce the sources of process errors and improve process latitude (the response of CD to an error). Interestingly, process latitude is similar to pitch resolution in that it also depends most strongly on the imaging wavelength and numerical aperture, though in more complicated ways.

Process latitude is an exceedingly general concept, but usually the most important process latitudes are the interrelated responses of CD to focus and exposure. Using the process window to characterize these responses (see Chapter 8), we define depth of focus (DOF) as *the range of focus which keeps the resist profile of a given feature within all specifications (linewidth and profile shape) over a specified exposure range*. One of the most insidious difficulties of lithography is that small features tend to have smaller depth of focus.

10.1.1 Defining Resolution

Resolution is, quite simply, the smallest feature that you are able to print (with a given process, tool set, etc.). The confusion comes from what is meant by ‘able’. For a researcher investigating a new process, ‘ability’ might mean shooting a number of wafers, painstakingly searching many spots on each wafer, and finding the one place where a small feature looks somewhat properly imaged. For a production engineer, the *manufacturable* resolution might be the smallest feature size which provides adequate yield for a device designed to work at that size. We can define resolution, similar to our definition of DOF, in such a way that it can meet these varied needs.

Producing an adequately resolved feature in a realistic working environment means printing the feature within specifications (linewidth, sidewall angle, resist loss, line edge roughness, etc.) over some expected range of process variations. As we have seen before, the two most common process variations are focus and exposure. Since our definition of depth of focus includes meeting all profile specifications over a set exposure range, a simple definition of resolution emerges: *the smallest feature of a given type which can be printed with a specified depth of focus*. This definition is perfectly general. If the exposure latitude specification used in the DOF definition is set to zero and the DOF specification in the resolution definition is set to zero, the ‘research’ use of the term resolution is obtained (if it prints once, it is resolved). If the exposure latitude and DOF specifications are made sufficiently large to handle all normal process errors encountered in a manufacturing line, the ‘manufacturing’ use of the term resolution is obtained. As with the

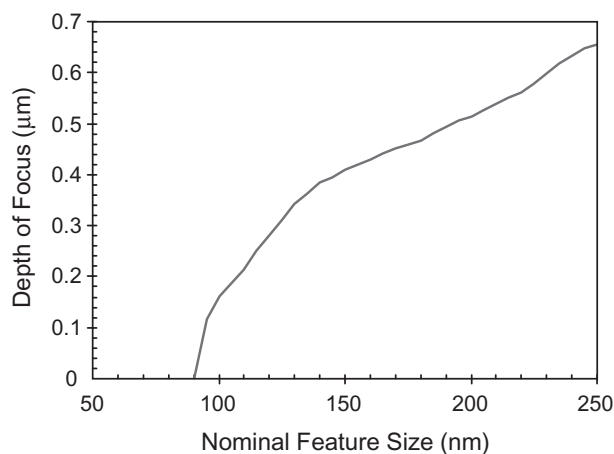


Figure 10.1 Resolution can be defined as the smallest feature which meets a given DOF specification. Shown are results for equal lines and spaces, $\lambda = 193\text{ nm}$, $NA = 0.9$, $\sigma = 0.7$, typical resist on a nonreflective substrate

definition of DOF, the choice of the specifications determines whether the resulting resolution is appropriate to a given application.

Figure 10.1 illustrates this concept of resolution. The depth of focus for a pattern of equal lines and spaces is shown as a function of feature size. (For this and subsequent figures, the DOF is based on profile specifications of $CD \pm 10\%$, sidewall angle $> 80^\circ$, resist loss $< 10\%$, and an exposure latitude specification of 6%. All focus and exposure errors are assumed to be random. Each data point assumes that nominal exposure and focus were adjusted to give the best process window and thus the largest possible DOF. Mask linearity – the ability to print different feature sizes at the same time – is not considered here, but could easily be added as a constraint.) If zero DOF is required, the equal line/space resolution for this process would be about 80 nm. A requirement of 160-nm DOF would increase the minimum printable feature size to 100 nm, and a requirement of 250-nm DOF would degrade the resolution further to 115 nm. Obviously, a simple statement of the resolution without clearly stating the DOF requirement (and thus the profile and exposure latitude requirements) would be of little use.

Figure 10.2 illustrates how a given process, tool set, etc., does not have a single resolution for all feature types. The contact hole shows the worst resolution under these conditions, while the isolated line has the greatest resolution for small required DOF. Figure 10.3 illustrates how a careful definition of resolution can elucidate fundamental lithographic behavior, such as the role of numerical aperture. For larger features, lower NA gives more depth of focus. But for smaller features, the DOF falls off more quickly for the lower NA. This results in the well-known effect of an optimum NA to give the greatest DOF. But it also impacts resolution in an interesting way. If no DOF is required, the resolution (the point where each curve in Figure 10.3 hits the x-axis) follows the familiar trend of increased resolution with increased NA. If, however, a nonzero DOF is required, the behavior of resolution with NA becomes more complicated.

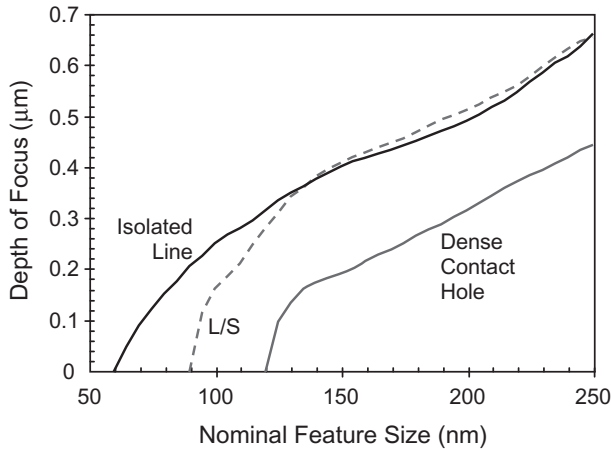


Figure 10.2 Comparison of the resolution for different feature types ($\lambda = 193 \text{ nm}$, $\text{NA} = 0.9$, $\sigma = 0.7$). Here, L/S means equal lines and spaces

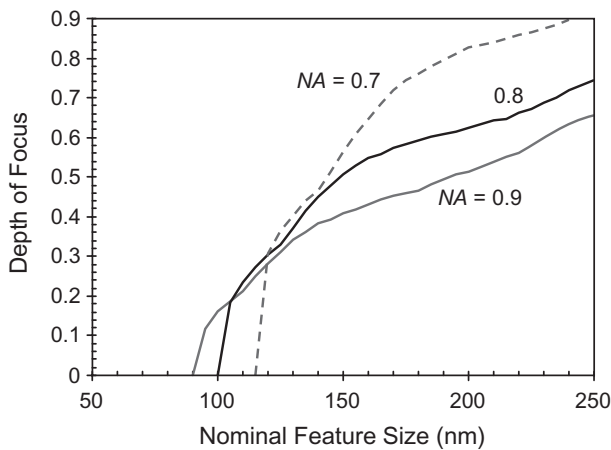


Figure 10.3 The definition of resolution can be used to study fundamental lithographic trends, such as the impact of numerical aperture (NA) on resolution ($\lambda = 193 \text{ nm}$, $\sigma = 0.7$, equal lines and spaces)

Figure 10.4 expands on the results of Figure 10.3 and shows the resolution of equal line/space arrays as a function of numerical aperture for different DOF specifications.² For example, with a required DOF of 200 nm, the resolution reaches an optimum (a minimum in the curve at a feature size of 105 nm) at a numerical aperture of 0.84. Larger numerical apertures actually reduce the resolution! As the required DOF is reduced, the NA which gives maximum resolution moves out to higher values. Also shown on the graph is the Rayleigh resolution criterion ($R = k_1\lambda/\text{NA}$) for comparison. For a nonzero

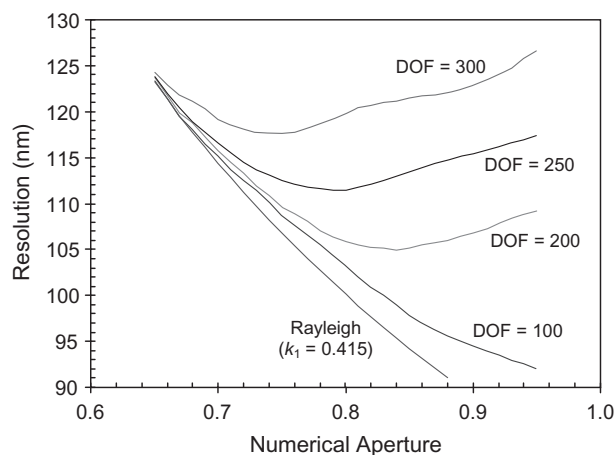


Figure 10.4 Resolution as a function of numerical aperture is more complicated than Rayleigh's criterion would imply ($\lambda = 193\text{ nm}$, $\sigma = 0.7$, equal lines and spaces). Graphs show different resolution versus NA results for different minimum DOF specifications in nanometers

required DOF, the Rayleigh criterion is not accurate in predicting the influence of NA on resolution.

10.1.2 Pitch Resolution

Although the above definition of resolution is perfectly general, there is a class of features where a slightly different approach is more appropriate. Consider a simple repeating mask pattern of equal lines and spaces. One could determine the resolution of this pattern type using the above definition, with the condition that the resist features must be equal lines and spaces as well (i.e. the desired linewidth equals the desired spacewidth). However, if we were to concentrate only on the line feature, we could easily overexpose our positive resist to produce a smaller linewidth and, most likely, a smaller 'resolution'. That is, keeping the pitch of the pattern (linewidth plus spacewidth) constant, we could improve the line feature resolution by a simple processing change, but only at the expense of the space feature resolution. Is this truly an improvement in resolution?

The answer depends on the application. If only the width of the line is critical, then resolution should be based only on the line feature. The electric performance of a device, for example, may be critically dependent on the linewidth of a given device structure, but only marginally affected by the accompanying space feature size. In most cases, however, the space feature is also critical (affecting capacitance at interconnect levels and strain at the gate level, for example). In fact, the ability to decrease both linewidth and spacewidth simultaneously allows manufacturers to shrink chip sizes, putting more chips on a wafer and providing a huge economic driver for the quest for better resolution. For such applications, where linewidth and spacewidth are both critical, one can modify the above definition of *feature* resolution to produce a definition for *pitch* resolution: *the smallest*

pitch of a given duty cycle which can be printed with a specified depth of focus, where duty cycle is defined as the ratio of spacewidth to linewidth.

At first glance the difference between the feature resolution and the pitch resolution seems almost trivial – hardly worth the effort to propose a separate definition. However, use of these two different types of resolutions reveals that the physical limits to resolution can be quite different for each. Consider the simple case of forming an image of an equal line/space mask pattern illuminated with a single wavelength, normally incident plane wave (i.e. coherent illumination). For such a case, there will be a hard cutoff for the pitch resolution: when the pitch drops below λ/NA no image whatsoever is formed for any duty cycle. Regardless of the profile, exposure latitude and DOF specifications, no pitch below this limit can be imaged. For the case of an isolated line, there is no equivalent ‘hard cutoff’ of the feature resolution, which instead exhibits a gradual reduction in profile control as the feature size is decreased.

In general terms, the feature resolution is limited by photoresist profile control and is a complicated function of wavelength and numerical aperture. Ultimately, the pitch resolution is limited by the cutoff of discrete diffraction information passing through the objective lens and is a relatively simple function of wavelength and numerical aperture. To understand this simple functionality, one must understand that a single diffraction order passing through the objective lens produces a single plane wave of light striking the wafer (see Chapter 2). Two plane waves at the wafer (coming from two separate diffraction orders) will interfere with each other to produce a sinusoidal pattern of light and dark (giving spaces and lines). If the two plane waves strike the image plane (i.e. the wafer plane) with angles θ_1 and θ_2 with respect to the optical axis (that is, a normal to the image plane), then the period of the resulting image will be

$$period = \frac{\lambda}{\sin \theta_1 - \sin \theta_2} \quad (10.2)$$

This general expression can be used to understand a variety of imaging situations. For the coherent illumination case described above, $\theta_2 = 0$ for the zero order and $\sin \theta_1$ has a maximum value of NA , giving the λ/NA resolution mentioned above. In other words, for three-beam imaging the minimum k_{pitch} is 1.0. For the special case of two symmetrical beams, $\theta_1 = -\theta_2 = \theta$ and the period becomes

$$period = \frac{\lambda}{2 \sin \theta} \quad (10.3)$$

The ultimate pitch resolution of an imaging tool is obtained when $\sin \theta$ becomes its maximum value, the NA :

$$ultimate \text{ pitch resolution} = \frac{\lambda}{2NA} \quad (10.4)$$

for all duty cycles. Thus, for two-beam imaging, the minimum k_{pitch} is 0.5. This ultimate pitch resolution uses two-beam imaging that can be obtained from an alternating phase-shifting mask or an optimized off-axis illumination scheme (see sections 10.3 and 10.4). However, the actual pitch resolution may not be this good if the resulting printed image

does not meet the required DOF specifications (that is, if the patterns are limited by the feature resolution of the line or the space making up the pitch).

10.1.3 Natural Resolutions

Besides the two kinds of resolution discussed so far, feature resolution and pitch resolution, a third type of resolution emerges from a study of optical imaging: natural resolutions of an imaging system. Since a discussion of natural resolution will make use of many of the concepts that will be developed in the sections that follow, its treatment will be delayed until the end of this chapter, in section 10.5.

10.1.4 Improving Resolution

Considering both pitch resolution and feature resolution, the following approaches are commonly used to improve resolution:

- Reducing the exposure wavelength
- Increasing the imaging lens numerical aperture
- Reducing k_{pitch} by using two-beam imaging
- Increasing the focus–exposure process window size
- Reducing the magnitude of process errors such as focus and exposure errors.

Unfortunately, several of these factors work against each other. For a given feature, there is an optimum numerical aperture that gives the largest process window. Increasing the NA further will reduce the process window, making the feature resolution worse even as the pitch resolution is improved. Reducing wavelength is always good, but as a practical matter, it is extraordinarily difficult since we are limited by our ability to engineer materials with the proper optical properties at the lower wavelength. Of course, reducing the magnitude of process errors is a never ending quest with cost being the only possible downside. That leaves two final resolution enhancement approaches: reducing k_{pitch} with two-beam imaging and increasing the size of the process window.

By far the most effective and popular process window improvement approach has been improvements in the photoresist. Over the years, resist capabilities have undergone dramatic progress. While we are still far from being able to ignore the photoresist (that is, we do not have a diffusionless infinite contrast resist), resist performance today is high enough that even small improvements in optical imaging can be seen in the final patterns.

Attempts to improve the process window by optical means (sometimes called optical ‘tricks’) include:

- Optimization of the mask pattern shape (called optical proximity correction, OPC)
- Optimization of the angles of light illuminating the mask (called off-axis illumination, OAI)
- Adding phase information to the mask in addition to intensity information (called phase-shifting masks, PSM).

Collectively, these optical approaches (each of which will be described in detail in the sections that follow) are known as resolution enhancement technologies (see Figure 10.5). While some techniques improve feature resolution at the expense of pitch resolution,

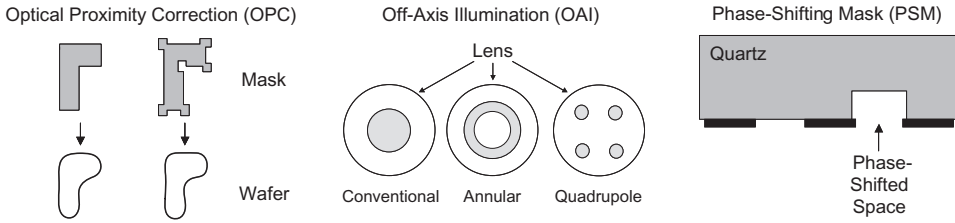


Figure 10.5 Examples of the three most common resolution enhancement technologies

many of the resolution enhancement technology (RET) approaches can improve pitch resolution and increase the process window simultaneously, a seemingly no-compromise path to resolution enhancement. However, the most promising RETs (especially the best PSM techniques) require a revolution in chip layout design that has yet to occur. Ultimately, a k_{pitch} as low as 0.5 is possible, but only for chips designed specifically to take advantage of these RETs.

10.2 Optical Proximity Correction (OPC)

Since, in general, the smallest features in a design are the hardest to print, lithography processes are optimized to best print those smallest features. Unfortunately, the larger features (which, while *larger*, cannot be considered large) are rarely faithfully reproduced using this same lithography process. The nonlinear nature of a high-resolution lithography process ensures that a ‘what you see is what you get’ imaging ideal, where all features on the mask are faithfully reproduced in resist, will never be achieved.

10.2.1 Proximity Effects

Proximity effects are the variations in the linewidth of a feature (or the shape for a 2D pattern) as a function of the proximity of other nearby features. The concept of proximity effects became prominent several decades ago when it was observed that electron beam lithography can exhibit extreme proximity effects (backscattered electrons can travel many microns, exposing photoresist at nearby features). Optical proximity effects refer to those proximity effects that occur during optical lithography (even though they may not be caused by optical phenomenon). The simplest example of an optical proximity effect is the difference in printed linewidth between an isolated line and a line in a dense array of equal lines and spaces, called the *iso-dense print bias*.

Although many factors may affect the iso-dense print bias, such as developer flow or PEB diffusion, in general this bias is fundamentally the result of optics – the aerial images for dense and isolated lines are different. For high-resolution features, the diffraction patterns from isolated and dense lines are significantly different (see Chapter 2), resulting in different aerial images, as shown in Figure 10.6. In this case, the isolated line will print wider than the dense line (assuming a positive photoresist), giving a positive iso-dense print bias. It is important to note that this result is not a ‘failing’ of the optical system,

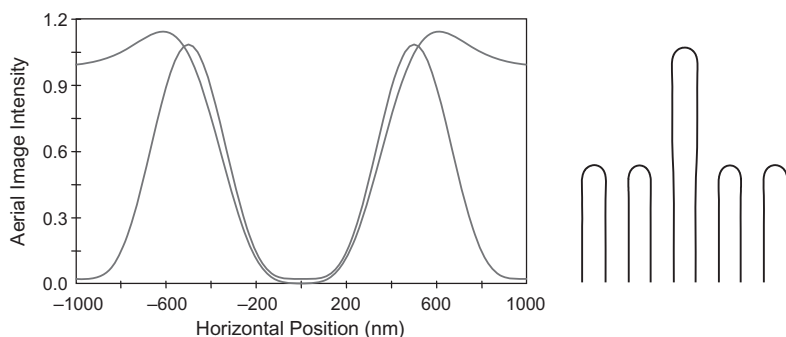


Figure 10.6 The iso-dense print bias is fundamentally a result of the difference in the aerial images between isolated and dense lines. In this case, the isolated line is wider than the line in a dense array of equal lines and spaces (0.5-micron features, $\lambda = 365$ nm, $NA = 0.52$, $\sigma = 0.5$)

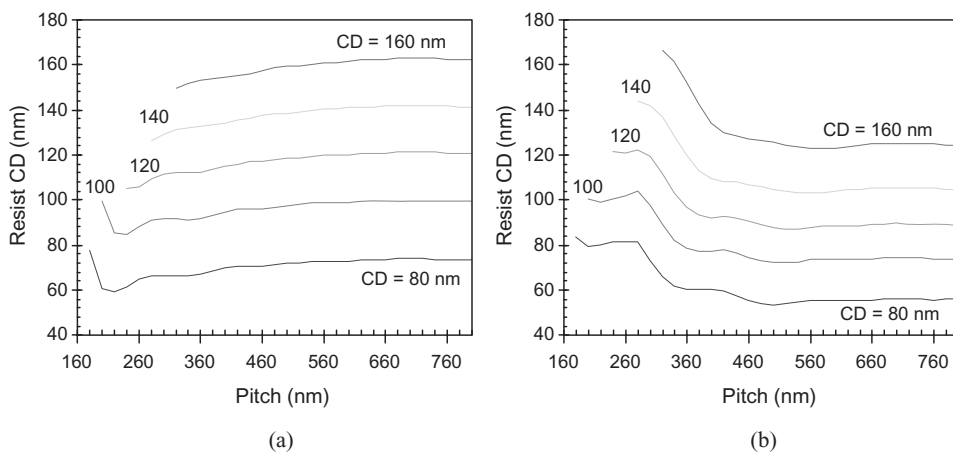


Figure 10.7 Resist CD through pitch for different nominal feature sizes (used to fully characterize 1D proximity effects) can be very different as a function of the optical imaging parameters used: (a) conventional illumination, $\sigma = 0.7$, and (b) quadrupole illumination, center $\sigma = 0.8$ ($\lambda = 193$ nm, $NA = 0.85$, binary mask, dose set to properly size the 100-nm line/space pattern)

but a natural consequence of the physics of imaging. Also, aberrations in the optical system can change the magnitude of the bias, sometimes significantly.

The proximity effect is very feature size dependent. For large features, the diffraction patterns for isolated and dense lines are similar, giving very little differences in the aerial images. As feature size shrinks, the differences grow. Thus, to fully characterize one-dimensional proximity effects, it is common to measure resist CD versus pitch for a range of nominal feature sizes (Figure 10.7). It is very important to note that the resulting

behavior (the CD-through-pitch curves) is very sensitive to the exact lithographic conditions used. For example, a switch from conventional to quadrupole illumination causes a dramatic change in the shape of the CD-through-pitch curves (Figure 10.7a vs. Figure 10.7b), even changing the sign of the iso-dense bias for some features.

The photoresist will also influence proximity effects. A systematic study using lithography simulation showed that the resist property that most significantly influences proximity effects is resist contrast.³ In simulation terminology, it is the resist dissolution selectivity parameter n of the original kinetic development model (which is directly proportional to resist contrast, see Chapter 7) that influences proximity effects. As an example of how an optical proximity effect might change with resist contrast, Figure 10.8 shows the influence of the next nearest feature on the linewidth of a nominal 400-nm feature using i-line lithography and a conventional resist. No proximity effects would result in the flat line shown at 400-nm linewidth. Five curves are shown corresponding to five different resist contrasts (subjectively called low, medium, high, state-of-the-art and future contrast resists). These resists correspond to dissolution selectivity parameters of 4, 5.5, 7, 10 and 16, respectively. The infinite contrast resist corresponds to the width of the image-in-resist itself since an infinite contrast resist will reproduce the image-in-resist exactly (barring diffusion).

Obviously, resist contrast plays an important role in determining the actual printed proximity effect. For example, the iso-dense bias (the difference in linewidth between isolated and dense features) for the state-of-the-art resist is twice that of the low-contrast resist for the case shown in Figure 10.8. Also, the lower-contrast resists show a dip in linewidth as the spacewidth is initially increased, whereas the higher-contrast resists do not. Incidentally, the range of printed linewidths often exceeds the iso-dense bias quite significantly, as Figure 10.8 clearly shows.

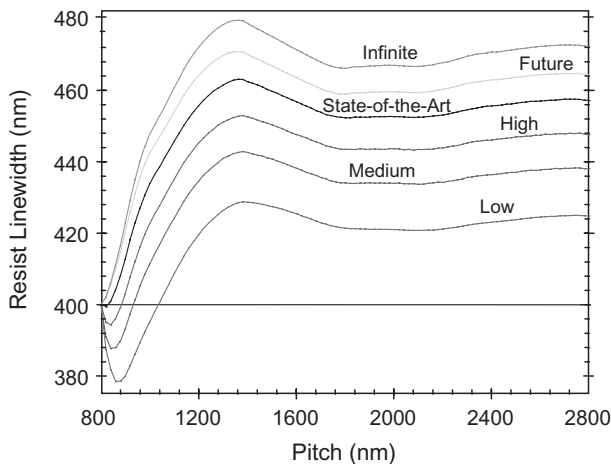


Figure 10.8 Proximity effects for different resist contrasts (400-nm nominal features, $NA = 0.52$, $\sigma = 0.5$, i-line). The increasing pitch corresponds to increasing distance between 400-nm lines

10.2.2 Proximity Correction – Rule Based

Since the difference between the desired and actual CD printed on the wafer as a function of feature size and pitch is, for the most part, a systematic error, it should be possible to correct for this error. The only practical means for correction is to change the CD on the mask to compensate for these proximity effects. This compensation is called *optical proximity correction* (OPC).

The initial goal of OPC, then, is to determine the optimal mask shape to get the desired resist shape – often called the ‘inverse problem’ in imaging. This simple question is more complicated than it appears. As with most such questions, the answer depends on what is meant by ‘optimal’. Consider first the simple case of printing long lines. What is the optimal width of the mask line to get the desired resist line for many different linewidths and proximity to other lines? One simple definition of optimum could be just obtaining the right linewidth at the nominal exposure dose and focus. As an example, consider a 100-nm resolution process using the 100-nm equal line/space pattern as the baseline. Without any bias on the mask for this feature, we determine the exposure dose to properly size this feature. Our simple requirement, then, is that all other mask features must properly print at their correct size at this dose as well. Since proximity effects prevent this from occurring naturally, we can only obtain this result by changing the feature sizes on the mask to correct for these proximity effects.

Figure 10.9 shows one example of what the mask bias solution might look like. The mask correction (CD bias) is defined as the actual absorber width that produces the desired resist CD minus the nominal (unbiased) absorber width. Thus, a positive bias means the absorber has been made bigger. Each curve shows the amount of linewidth bias needed as a function of the nominal feature width. For this example, we have defined our starting point as zero bias for the most difficult feature, the 100-nm lines and spaces. An alternate

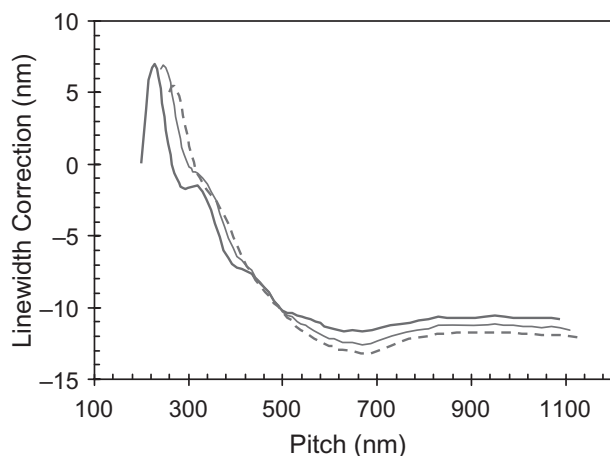


Figure 10.9 Design curves of the mask linewidth bias (in wafer dimensions) required to make all of these features print at the nominal linewidth: 100 nm (thick line), 120 nm (thin line) and 140 nm (dashed line). Dose set to require no bias at 100-nm lines and spaces ($\lambda = 193$ nm, $NA = 0.93$, $\sigma = 0.7$, 6 % ESPM)

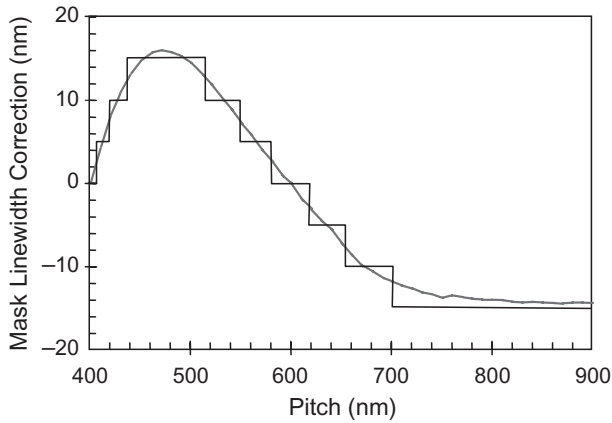


Figure 10.10 Discretized design curve (the stair-step approximation to the actual smooth curve) appropriate for use in a design rule table (5-nm correction grid used)

(though less common) approach would be to set zero bias for some large feature and adjust the bias of the smaller features to match the dose-to-size of that larger feature.

Our definition of the optimum mask bias so far is a simple one – the bias which gives the proper printed linewidth at the nominal process conditions. Other definitions may include the tolerance to variations in process conditions (for example, the maximum overlap of the focus–exposure process windows of the various features). Ultimately, the best solution is that which gives the tightest distribution of linewidths on the wafer for all of the features in the presence of typical process variations.

While the curves of Figure 10.9 show how mask correction is a continuous function of nominal CD and pitch, the actual implementation of these corrections on a real design is often snapped to a design grid in order to ease the design and mask-making burden. Typical design grids are 1–3 % of the minimum resist feature size. For example, a typical design curve is snapped to a 5-nm grid in Figure 10.10. When discretized in this way, the information in the design curve can be easily captured in a correction table, called an OPC rules table.

While all of the examples shown here have been symmetrical (the proximity of other features is identical on both sides of the main feature), in general this will not be the case. Thus, corrections are better described as edge movements rather than linewidth changes. A simple OPC rule might be something like this: ‘If a feature is between A and B in width, and the next feature to the right is between X and Y away, move the right edge of the feature by Z’. Thus, the shape of the feature in the design is changed (its edge is moved) depending on the size of the feature and its proximity to its nearest neighbor. The values of the parameters of the rule (the amount of correction Z as a function of A, B, X and Y in this example) are empirically determined for a given lithography process. Since a large set of these parameters is typically required for each rule, the collection of these parameters is put into a table called a rules table. Figure 10.11 shows an extremely simple case where one-dimensional edge movements based on a rules table were applied.

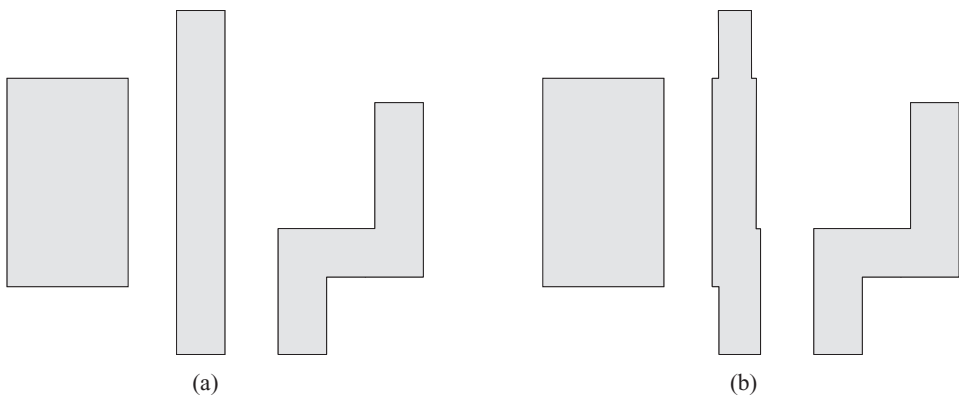


Figure 10.11 A small section of a design (a) before, and (b) after correction of the middle feature with a simple 1D rule-based correction

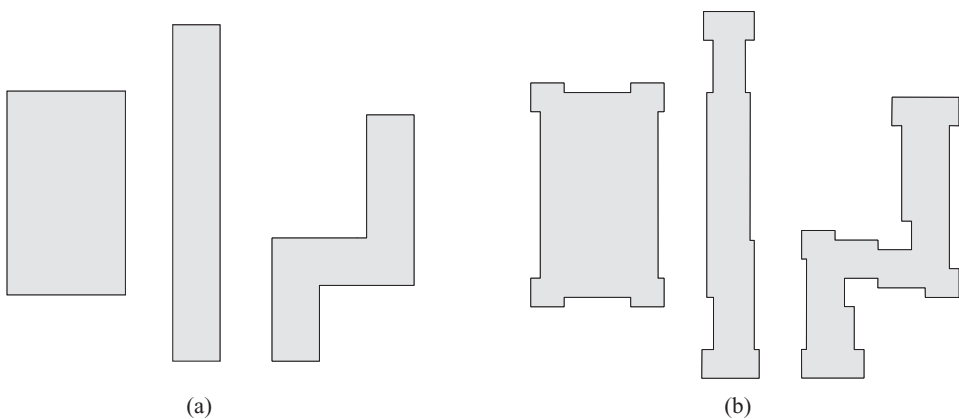


Figure 10.12 A small section of a design (a) before, and (b) after the use of a simple 1.5D rule-based correction

There are two key aspects of successfully implementing rule-based OPC: (1) having the right set of rules, and (2) having the right parameter values for those rules. While rule-based OPC is conceptually simple and easy to implement for one-dimensional corrections, the rules can get very cumbersome when considering two-dimensional effects. Corner rounding and line-end shortening (see Chapter 8) are two particular examples of important 2D proximity effects that should be addressed with OPC. An intermediate approach is to treat line-ends with their own set of rules, apply serifs to corners, and use 1D rules for all of the other edges. Sometimes called ‘1.5D’ correction, an example of this approach is shown in Figure 10.12.

Rule-based OPC is fairly simple to implement. First, a set of CD-through-pitch curves are experimentally measured for a given process. From these data, a one-dimensional rules table is created by interpolating from these data. Optimum line-end treatments and

corner serif sizes are also empirically determined for the given process. Corrections are made by implementing the rules as scripts for a design rule checker (DRC) software package. This approach to OPC became important when feature sizes dropped below 500nm, and was widely used at the 250-nm technology generation. However, by the 130-nm technology generation, the accuracy and robustness of rules-based OPC began to falter. It was at this technology generation that most semiconductor manufacturers began to switch to model-based OPC.

10.2.3 Proximity Correction – Model Based

Rule-based OPC suffers from a difficult scalability problem: in order to increase the accuracy of the OPC, the number of rules must grow. This growth in rules is highly non-linear – a small increase in OPC accuracy may require a very large increase in the number of rules. Additionally, the required accuracy of OPC scales at least linearly with the technology node dimensions. Since each rule requires several experimentally derived parameters, the result is a dramatic increase in effort. For these reasons, most semiconductor manufacturers realized that *model-based OPC* was required at the 90-nm technology generation.

Model-based OPC replaces experimentally derived rules with a calibrated lithography model that predicts the proximity effects for the actual chip pattern. Edges of the features are then iteratively moved until the predicted resist shape matches the desired feature shape to within a preset tolerance. A well-designed model-based OPC procedure requires only very few rules to operate. Instead, considerable effort is put into the development and calibration of a simplified lithography model that will predict how each design pattern will print.

The constraints on the lithography models used for OPC are severe. Since an entire chip must be simulated (with its hundreds of millions or even billions of features), OPC models must be very fast and should also be parallelizable (to take advantage of distributed computing). At the same time, these models must be very accurate, predicting the printed shapes on the wafer for every pattern of the design accurately to within a few percent of the minimum design rule size. This combination of very fast speed with acceptable accuracy has resulted in the development of hybrid physical/empirical lithography models, sometimes called *compact models*.

OPC models combine a reasonably accurate and physically correct aerial imaging model with a very simple resist model. Etch effects are sometimes included, or separate etch models are applied, to predict the after-etch feature sizes and shapes. For the aerial image calculation, an approximate solution to the Hopkins imaging equations called *Sum of Coherent Sources* (SOCS) provides extremely fast computation times by pre-computing a small number of coherent convolution kernels for a given source and lens pupil (see Chapter 2). Diffusion effects in the resist are usually approximated by allowing the aerial image to diffuse, which can be conveniently calculated using a Gaussian diffusion kernel in the SOCS formulation. Resist development effects are approximated using a threshold plus bias or a variable threshold model (see Chapters 2 and 7). Alternatively, resist effects can be captured as empirical kernels in the SOCS formulation. In either case, these empirical resist models require extensive calibration for a given resist process.

Model-based OPC begins by first dividing the original design into edge segments, each of which can be individually moved during the OPC process. Simulation of the design predicts a resist shape, which is then compared to the original design at various measurement points (for example, the midpoint of the segment) to calculate an edge placement error (EPE, see Chapter 8). This EPE is then used to guess how much the segment must be moved so that the resist shape will have zero EPE. Edge segments are iteratively moved and the resist shape resimulated until all of the EPEs are below a preset limit (typically requiring three to six iterations). As with rule-based OPC, edge positions are usually snapped to a design grid to reduce the design and mask complexity. An example of model-based OPC is shown in Figure 10.13. The aggressiveness of the OPC can be adjusted by controlling the segment size (smaller segments mean more aggressive OPC), the smallest movement allowed for an edge compared to its connecting edge (that is, the smallest allowed *jog* size), and the design grid, among other parameters. A more aggressive OPC results in a larger number of design vertices and greater mask-making cost.

In general, model-based OPC is used to find a mask shape that produces an acceptable resist shape for a given lithography process that is assumed to be operating at its nominal conditions, i.e. at best focus and exposure. Thus, while OPC can be used to extend the

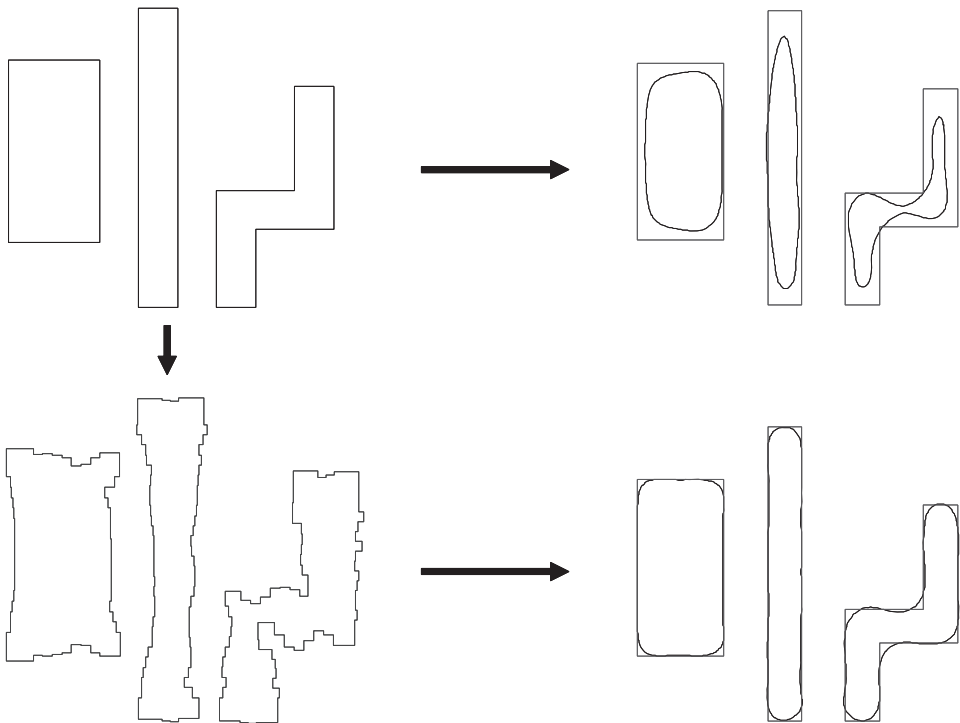


Figure 10.13 Example of model-based OPC: the original design (upper left) prints very poorly (upper right). After aggressive model-based OPC, the resulting design (lower left) prints very close to the desired shape (lower right). OPC and simulations done using PROLITH

linear resolution (the smallest features that can be printed while still printing all larger features acceptably), it does not address the issue of process window, and thus the true manufacturable resolution. One important technique used with OPC that addresses process window as well as sizing issues is the *subresolution assist feature*, as described next.

10.2.4 Subresolution Assist Features (SRAFs)

Scattering bars, also called subresolution assist features (SRAFs), are narrow lines or spaces placed adjacent to a primary feature in order to make a relatively isolated primary line behave lithographically more like a dense line.⁴ The problem being solved is generically described as the problem of iso-dense bias. Isolated features will almost always print at a feature size significantly different than the same mask feature surrounded by other features. The pitch curves of printed CD versus pitch for various nominal mask dimensions show the problem (see Figure 10.7). While sizing the mask to give the correct CD on the wafer for all pitches certainly works (this is the conventional OPC approach), there is another isolated-versus-dense difference that is not addressed by this bias OPC.

The response of an isolated feature to focus and exposure errors is significantly different than the same-sized dense line. Figure 10.14 shows example focus–exposure matrices for dense and isolated lines after the isolated line has been sized to give the proper CD at the best focus and exposure needed by the dense features. The different shapes of the Bossung curves produce different shapes for the process windows, which limits the overlapping depth of focus even when the features nominally have the same best exposure dose.

Scattering bars are designed to reduce the difference in the focus response of an isolated feature compared to a dense feature by making the isolated feature seem more ‘dense’. This becomes especially important when an off-axis illumination scheme is optimized for greatest depth of focus of the dense features (a topic that will be discussed extensively in section 10.3). The overlapping process window for the dense and isolated lines of Figure 10.14 is shown in Figure 10.15a. The curvature of the isolated process window severely limits the useable, overlapping DOF.

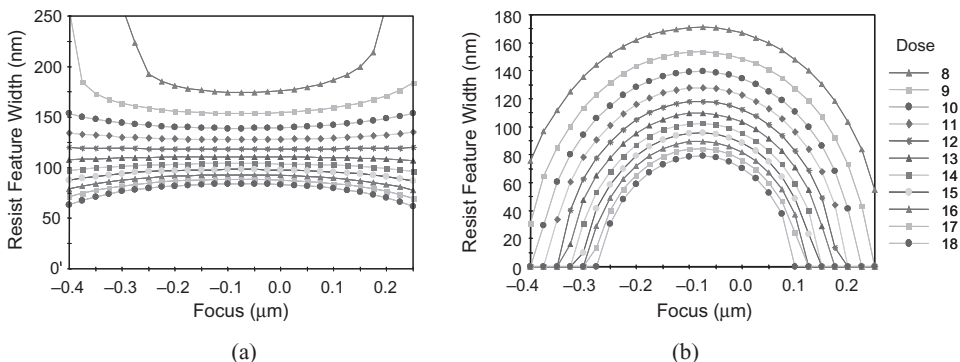


Figure 10.14 Focus–exposure matrices (Bossung curves) for (a) dense and (b) isolated 130-nm features (isolated lines biased to give the proper linewidth at the best focus and exposure of the dense lines, $\lambda = 248$ nm, NA = 0.85, quadrupole illumination optimized for a 260-nm pitch)

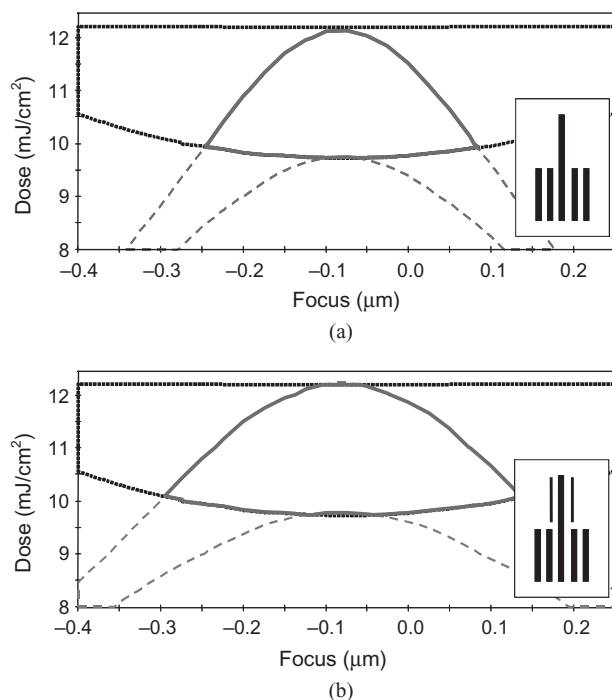


Figure 10.15 Overlapping process windows generated from the focus–exposure matrices of dense and isolated lines for (a) isolated lines with bias OPC (overlapping DOF = 300 nm) and (b) isolated lines with scattering bars (overlapping DOF = 400 nm)

An SRAF, as the name implies, is a subresolution feature that is not meant to print. In fact, it must be carefully adjusted in size so that it never prints over the needed process window. This determines the most important trade-off in scattering bar design: make the assist features as large as possible in order to create a more dense-like mask pattern, but not so large as to print. Generally, these assist features are centered on the same pitch for which the off-axis illumination was optimized, though a more careful design will optimize their position and size to maximize the improvement in overlapping process window. As a result, the use of assist features allows the lithographer to design an off-axis illumination process optimized for dense patterns that can also be used to print more isolated features.

The assist bars used in Figure 10.15b were 50 nm ($k_1 = 0.17$) in size (wafer dimensions), and resulted in an increase in the overlapping DOF from 300 nm, when only bias OPC was used, to 400 nm. Further improvement can be obtained by using ‘double’ scattering bars, where a second set of scattering bars is placed further away to create an effective five-bar pattern. Of course, this requires enough free space around the primary feature to actually be able to fit these extra assist features.

While the concept of using scattering bars to improve the DOF of isolated features is a simple one, its practical implementation is anything but simple. Unlike the idealized case of an isolated line, real patterns contain lines with a variety of pitches (i.e. nearby

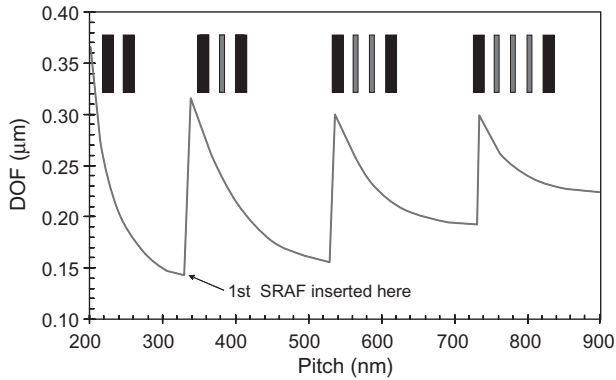


Figure 10.16 Schematic diagram of SRAF placement showing the discontinuous effect of adding an SRAF as the pitch grows (main feature size is 100 nm)

patterns with different distances away), each of which must be outfitted with an optimal assist feature or features, if one can fit. While bias OPC can be used on the intermediate cases where the space between two lines is not large enough to accommodate an assist feature, these intermediate pitches do not benefit from the DOF advantages of SRAFs (see Figure 10.16). And then, of course, there is the problem of what to do with line ends, corners, and other 2D patterns. Rule-based SRAF placement is quite common, but has difficulty with 2D placement. Model-based SRAF placement is difficult, but shows promise for complex 2D geometries. These issues can be resolved, however, and subresolution assist features are commonly used in many chip designs. Polysilicon gate and contact levels, in particular, have seen benefits from using SRAFs. For contacts and other dark-field mask levels, the SRAFs take the form of clear slots (spaces) rather than assist lines.

10.3 Off-Axis Illumination (OAI)

Off-axis illumination^{5,6} (OAI) is one of the three major resolution enhancement technologies that have enabled optical lithography to push practical resolution limits far beyond what was once thought possible. In order to effectively use off-axis illumination, the shape and size of the illumination must be optimized for the specific mask pattern being printed. This section will describe how to optimize the most popular types of off-axis illumination – dipole, quadrupole and annular illumination – to maximize depth of focus for a given pitch.

Off-axis illumination refers to any illumination shape that significantly reduces or eliminates the ‘on-axis’ component of the illumination, that is, the light striking the mask at near-normal incidence. By tilting the illumination away from normal incidence, the diffraction pattern of the mask is shifted within the objective lens. For the case of a repeating pattern, the diffraction pattern is made up of discrete diffraction orders. If the pitch of the repeating pattern is small, only a few diffraction orders can actually make it

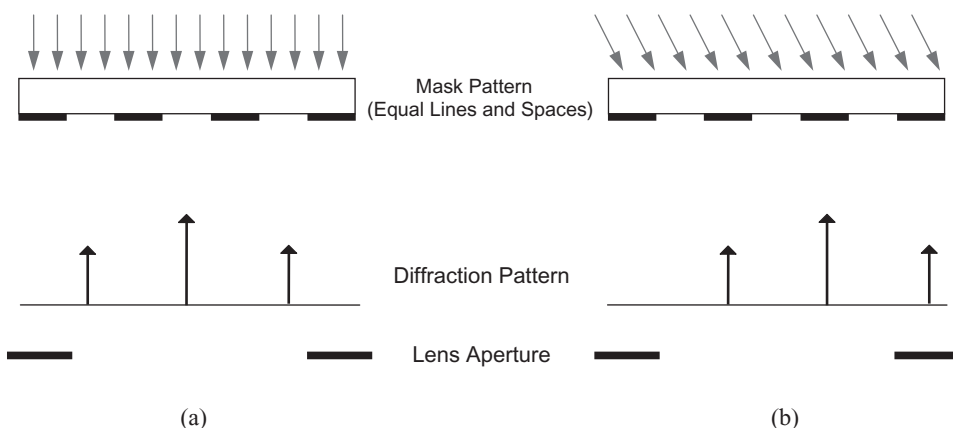


Figure 10.17 Off-axis illumination modifies the conventional imaging of a binary mask shown in (a) by tilting the illumination, causing a shift in the diffraction pattern as shown in (b). By positioning the shifted diffraction orders to be evenly spaced about the center of the lens, optimum depth of focus is obtained

through the finite size lens. As discussed in Chapter 3, placing those diffraction orders that make it through the lens evenly about the center of the lens leads to improved depth of focus (DOF). Thus, the main advantage of off-axis illumination is an increase in DOF (and thus the resolution) for small-pitch patterns. For these small pitch patterns, OAI changes the imaging from three-beam to two-beam imaging.

In spatial frequency terms, the distance between the zero and first diffracted orders is $1/p$, where p is the pitch. It will be convenient to convert this spatial frequency distance to ‘sigma space’, the spatial frequency normalized by λ/NA . In this normalized coordinate system, the maximum spatial frequency passing through the lens is 1.0. Thus, the distance between diffraction orders in sigma space is $\lambda/(pNA)$. To center the zero and first orders about the center of the lens, the zero order (found at the exact center of the lens for normally incident light) must be shifted by $\lambda/(2pNA)$ in sigma space. Thus, this becomes the optimum illumination tilt to give maximum DOF (Figure 10.17). Of course, tilting in the opposite direction [a $-\lambda/(2pNA)$ shift in sigma space] will produce the same effect. Combining both tilts into one illumination shape produces an illumination called *dipole illumination* that adds the required telecentricity behavior to the imaging. Note that the optimum illumination tilt is pitch dependent.

Real lithography, however, adds a significant complication to this otherwise simple picture. The line/space pattern shown in Figure 10.17 has a specific orientation (the lines are running into and out of the page) that results in an optimum tilt as shown in the figure. Most integrated circuit designs will contain many line and space-like features that are oriented both vertically and horizontally. A perspective plot of the same diffraction situation may make this point clearer, as shown in Figure 10.18. If the illumination is tilted by the amount discussed above, that tilt, in a specific direction, will only help the lines and spaces that are properly oriented with respect to that tilt. The other orientation of lines will not only not be improved by the illumination tilt, they are likely to

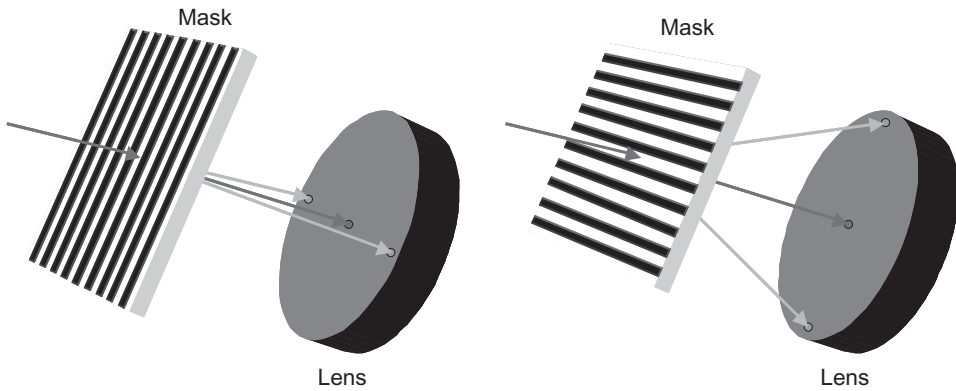


Figure 10.18 The position within the lens of the diffracted orders from a pattern of lines and spaces is a function of the orientation of the lines and spaces on the mask

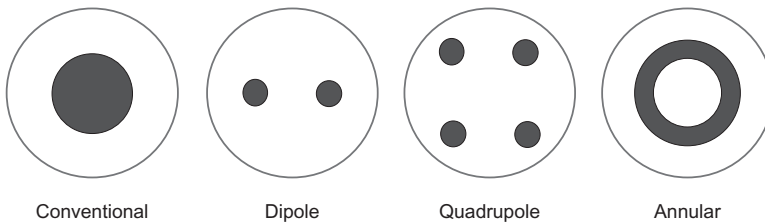


Figure 10.19 Various shapes for conventional and off-axis illumination

be significantly degraded in imaging performance. If both vertical and horizontal lines are to be imaged together on the same mask, an illumination shape must be used that provides optimum tilts for both geometries. The simplest shape that provides this optimum tilt for both horizontal and vertical line/space patterns is called *quadrupole illumination*.

Quadrupole illumination takes the optimum dipole generated for one orientation of lines and spaces, then shifts it both up and down in the other direction to create the proper angles for the other orientation of lines. The result is four poles evenly spaced about the center of the lens, as shown in Figure 10.19. In sigma space, the radial position of the center of each pole with respect to the center of the lens that gives optimum DOF is $\sqrt{2}\lambda/(2pNA)$. Note that this positioning of the quadrupoles gives the same horizontal and vertical spacing between poles as in the dipole case, but places them closer to the edge of the lens aperture.

While the quadrupole shape provides optimal performance for vertical and horizontal lines, other orientations (such as a line/space array oriented at 45°) will not be optimum. For any orientation of lines, the optimal dipole for that pattern will be spread in a direction perpendicular to the line orientation, and can be shifted parallel to the lines in any amount that keeps the dipoles within the lens. If the mask will contain arbitrary orientations of lines, many rotations of the dipoles will produce an annulus of illumination (and

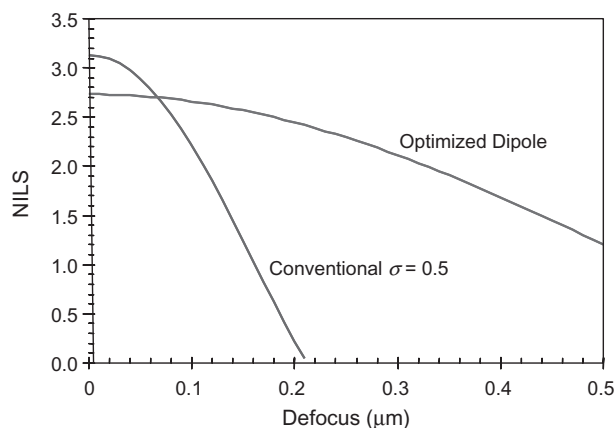


Figure 10.20 The impact of off-axis illumination on the log-slope defocus curve ($NA = 0.85$, $\lambda = 193\text{ nm}$, binary chrome-on-glass mask, 120-nm lines and spaces). The dipole had a radius of 0.2 in sigma space

thus is called *annular illumination*). The optimum center of the annular ring is the same as the optimum dipole position.

By switching from three-beam imaging to two-beam imaging, OAI improves the through-focus performance of the image, but degrades the in-focus quality of the image. Since one of the diffraction orders is lost when the illumination is tilted, the in-focus image will have reduced image log-slope (that is, reduced NILS). But if the tilt is optimized for the pitch being imaged, this reduced NILS will not change much as the image goes out of focus, as seen in Figure 10.20.

For each illumination shape discussed – dipole, quadrupole and annular illumination – there is one illumination size that maximizes the DOF for a given pitch. However, this illumination shape is only optimum for that one pitch. While pitches close to this optimum will get most of the benefit of the off-axis illumination, pitches sufficiently far away from the optimum will receive little or no benefit. In particular, isolated lines do not see the benefit of improved DOF when using off-axis illumination. Figures 10.21 and 10.22 show an example of how NILS and the DOF of a 100-nm line vary with pitch for quadrupole illumination nominally optimized for a 200-nm pitch. The DOF reaches a maximum when the pitch is near the designed-for pitch of 200 nm (the nonzero diameter of the poles means that the maximum DOF is obtained at a slightly larger pitch, in this case 230 nm). As the pitch increases, the DOF very quickly drops, leveling off at the DOF of an isolated line (which in this case is about half of the maximum DOF).

There is, however, a very convenient solution to the problem of lack of DOF for isolated lines: subresolution assist features. As discussed before, adding assist features around the more isolated lines will make them behave more like dense features. When using OAI, this means that the more isolated features can gain some of the benefit that OAI gives in terms of DOF. But, as we have seen before, SRAFs can be inserted between lines only if there is sufficient room to fit them. In general, SRAFs can be inserted when the pitch is greater than about 1.7 times the minimum pitch. For pitches less than about 1.3 times

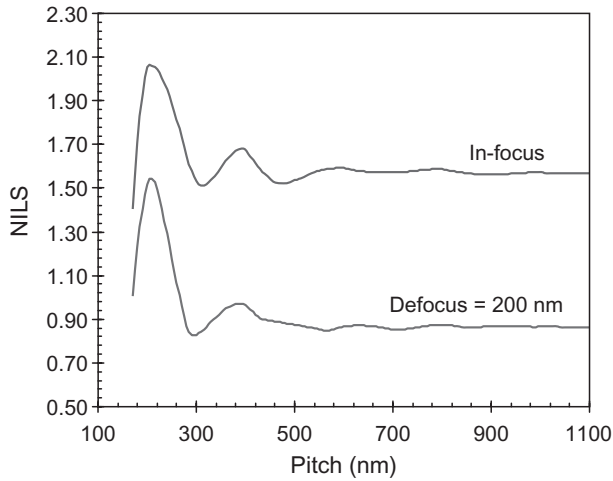


Figure 10.21 Quadrupole illumination optimized for a pitch of 200 nm showing how NILS varies with pitch both in-focus and with a moderate amount of defocus ($NA = 0.85$, $\lambda = 193$ nm, 100-nm line, chrome-on-glass mask, quadrupole settings of 0.8/0.2)

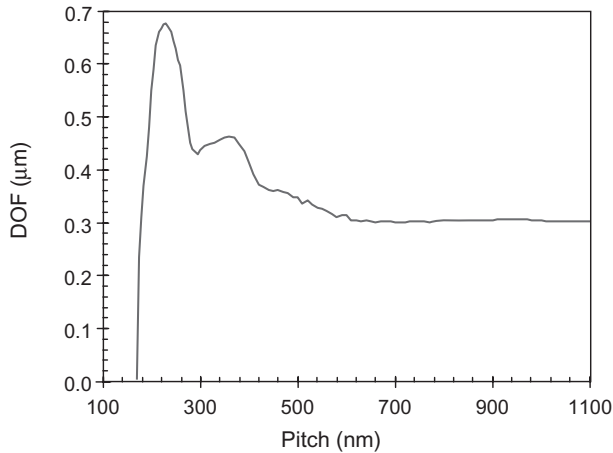


Figure 10.22 Quadrupole illumination optimized for a pitch of 200 nm showing how isolated lines do not show improved DOF ($NA = 0.85$, $\lambda = 193$ nm, 100-nm line, chrome-on-glass mask, quadrupole settings of 0.8/0.2)

the minimum pitch, the off-axis illumination provides sufficient DOF. Thus, a range of pitches emerges, roughly about 1.3–1.7 times the minimum pitch, where NILS and DOF are low and there is no possibility of using SRAFs to improve DOF. These pitches are sometimes called ‘forbidden’ pitches,⁷ indicating the lithographer’s desire that these pitches be avoided during circuit design. The forbidden pitch phenomenon significantly complicates the use of off-axis illumination since most chip designs will employ a wide range of pitches.

The three major types of off-axis illumination – dipole, quadrupole and annular illumination – have been described above. There are, however, a wide range of other shapes possible. Like line/space patterns, any repeating two-dimensional pattern will have an optimum source shape to maximize image quality over a range of focus. Thus, customized source shapes are often used whenever a certain repeating 2D shape is the most critical pattern in the design (commonly the case for DRAM manufacturing).

10.4 Phase-Shifting Masks (PSM)

For a conventional chrome-on-glass mask, the idealized mask transmittance is considered to be binary: the light is 100 % transmitted through the glass areas and 100 % blocked by the chrome. The resulting Kirchhoff approximation predicts a mask (electric field) transmittance function $t_m(x,y)$ that is either 0 or 1 (and thus the term ‘binary’ mask to describe this type of transmission). A *phase-shifting mask* (PSM) modifies not only the amplitude of the transmitted electric field but the phase of the transmitted light as well.

Consider the cross section of a mask structure shown in Figure 10.23. Two nearby regions of the mask transmit 100 % of the monochromatic light, but experience different optical path lengths. In one region, the light passes through extra glass (or fused silica) of thickness d and refractive index n_g . In a nearby region, the light travels through air of the same thickness (with refractive index of 1). The phase difference between two plane waves traveling perpendicularly through the mask will be

$$\Delta\phi = 2\pi d(n_g - 1)/\lambda \quad (10.5)$$

By adjusting the thickness of this extra layer of glass, any phase difference between the two waves can be obtained. As we shall soon see, it will be very advantageous to make this phase-shift exactly 180° (π). The thickness required to achieve this will be

$$d_{180} = \frac{\lambda}{2(n_g - 1)} \quad (10.6)$$

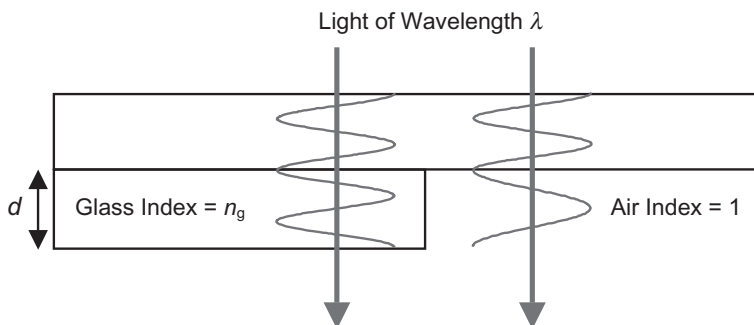


Figure 10.23 Cross section of a mask showing how the phase of the light transmitted through one part of the mask can be shifted relative to the phase of light transmitted through a nearby part of the mask

The fused silica most commonly used for mask blanks has a refractive index of 1.56 at 193 nm and 1.51 at 248 nm. Thus, the resulting glass thickness to give a π phase shift is slightly less than one wavelength.

The phase shift depicted in Figure 10.23 can be achieved by adding extra fused silica to the region on the left, or by etching away the same amount of mask blank material on the right. Alternately, some other material might be added to the mask blank so that both the phase and the amplitude transmittance of the light can be manipulated. There are an extremely large number of possible variations when both the amplitude and the phase of the transmitted light can be varied. However, practical mask-making considerations limit the number of different types of transmittance regions on the mask to a small number, typically two or three. And while many styles of phase-shifting masks have been investigated, only the two most common, alternating PSM and attenuated PSM, will be described in some detail below.

10.4.1 Alternating PSM

A conventional, binary chrome-on-glass mask of lines and spaces will produce a diffraction pattern of discrete diffraction orders at spatial frequencies that are multiples of one over the pitch (see Chapter 2). For a high-resolution pattern, only the zero and the plus and minus first diffraction orders pass through the lens (which has a spatial frequency cutoff of NA/λ), as seen in Figure 10.24a. In fact, it is the interference of the zero-order light with the first orders that produces the bright and dark image of the proper pitch. If the pitch is made too small, the first-order light diffracts at an angle too large to fit through the objective lens and no image is produced. The resolution limit, then, occurs when the

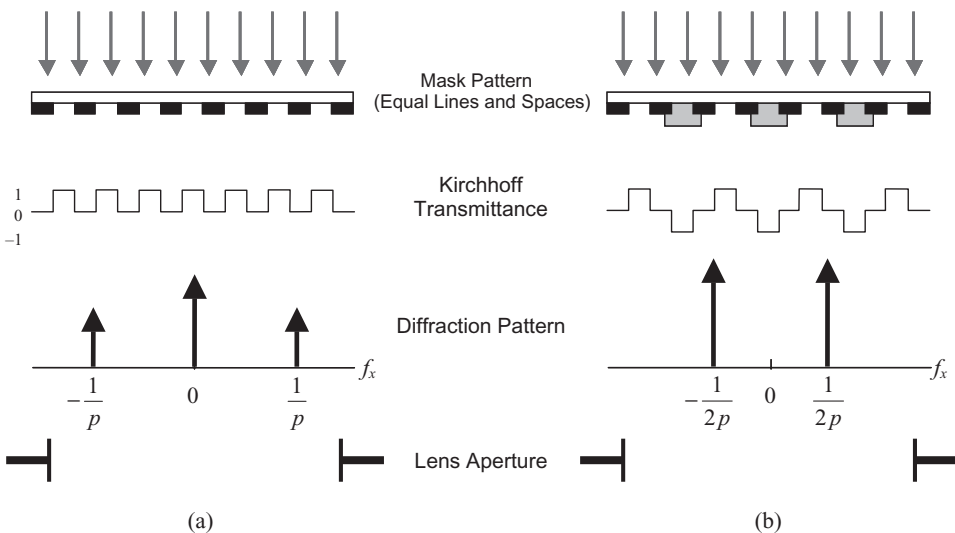


Figure 10.24 A mask pattern of equal lines and spaces of pitch p showing the idealized amplitude transmittance function and diffraction pattern for: (a) binary chrome-on-glass mask; and (b) alternating phase-shift mask

first diffracted order (spatial frequency of $1/\text{pitch}$) lands exactly at the edge of the aperture (spatial frequency of NA/λ) so that the minimum resolvable pitch is equal to λ/NA (i.e. $k_{\text{pitch}} = 1.0$). Additionally, the use of partially coherent illumination can extend this classical resolution limit, but only at the expense of reduced image quality.

Consider now a repeating line/space pattern (with lithographic pitch $p = \text{linewidth} + \text{spacewidth}$) where every other space has been phase-shifted by 180° . Letting w be the spacewidth, the diffraction pattern can be calculated as the superposition of one repeating space with pitch $2p$ and transmittance 1, plus another repeating space of pitch $2p$ and transmittance -1 , shifted by p . Applying the superposition and the shifting theorems of the Fourier transform, the diffraction pattern will be

$$\begin{aligned} T_m(f_x) &= \sum_{j=-\infty}^{\infty} \frac{\sin(j\pi w/2p)}{j\pi} \delta\left(f_x - \frac{j}{2p}\right) - \sum_{j=-\infty}^{\infty} e^{i2\pi f_x p} \frac{\sin(j\pi w/2p)}{j\pi} \delta\left(f_x - \frac{j}{2p}\right) \\ &= \sum_{j=-\infty}^{\infty} \frac{\sin(j\pi w/2p)}{j\pi} [1 - (-1)^j] \delta\left(f_x - \frac{j}{2p}\right) \end{aligned} \quad (10.7)$$

When j is even, the diffraction order from the shifted space exactly cancels out the diffraction order from the unshifted space. Thus, only the odd orders survive, giving

$$T_m(f_x) = \sum_{j=-\infty}^{\infty} a_j \delta\left(f_x - \frac{j}{2p}\right) \quad (10.8)$$

where

$$a_j = \begin{cases} 0, & j = \text{even} \\ \frac{2 \sin(j\pi w/2p)}{j\pi}, & j = \text{odd} \end{cases}$$

The resulting mask is called an *alternating phase-shift mask*⁸ (also called a Levenson PSM or abbreviated Alt-PSM), as depicted in Figure 10.24b. For small pitch patterns, the image is obtained from the interference of the two first diffraction orders, located at the spatial frequencies of $\pm 1/2p$. For coherent TE illumination, the image is

$$I(x) = 2a_1^2(1 + \cos(2\pi x/p)), \quad a_1 = \frac{2 \sin(\pi w/2p)}{\pi} \quad (10.9)$$

The resolution limit is again obtained when these first diffracted orders just barely pass through the edge of the lens, making the minimum resolvable pitch equal to $0.5\lambda/NA$. Thus, the use of an alternating phase-shift mask leads to two-beam imaging with its potential for improved resolution and better DOF. Unlike off-axis illumination, however, the resulting diffraction pattern provides optimum defocus performance for all pitches, since the two diffraction orders will always be equally spaced about the center of the lens, regardless of pitch. Also, the two orders will always be of equal magnitude, a condition that results in the maximum image log-slope for two-beam imaging.

While the example given above is for a repeating pattern of lines and spaces, alternating PSM can be applied to the printing of any narrow, dark feature by shifting the phase of light on one side of the feature compared to the other side. Consider an isolated line of

width w , where the clear region on the right side of the feature is phase-shifted by 180° relative to the transmittance on the left side of the feature (Figure 10.25). The diffraction pattern for this phase-shifted, isolated line is

$$T_m(f_x) = i \frac{\cos(\pi f_x w)}{\pi f_x} \quad (10.10)$$

For coherent TE illumination, the resulting aerial image can be expressed in terms of sine integrals:

$$I(x) = \frac{1}{\pi^2} \left[\text{Si} \left(\frac{2\pi NA}{\lambda} (x + w/2) \right) + \text{Si} \left(\frac{2\pi NA}{\lambda} (x - w/2) \right) \right]^2 \quad (10.11)$$

where

$$\text{Si}(\theta) = \int_0^\theta \frac{\sin z}{z} dz$$

It is interesting to examine the behavior of this image as w goes to zero. For an unshifted line, of course, the image would disappear as the chrome linewidth went to zero. But as Figure 10.26 shows, as the chrome width goes to zero the aerial image width approaches



Figure 10.25 Cross section of a mask of an isolated phase-shifted line

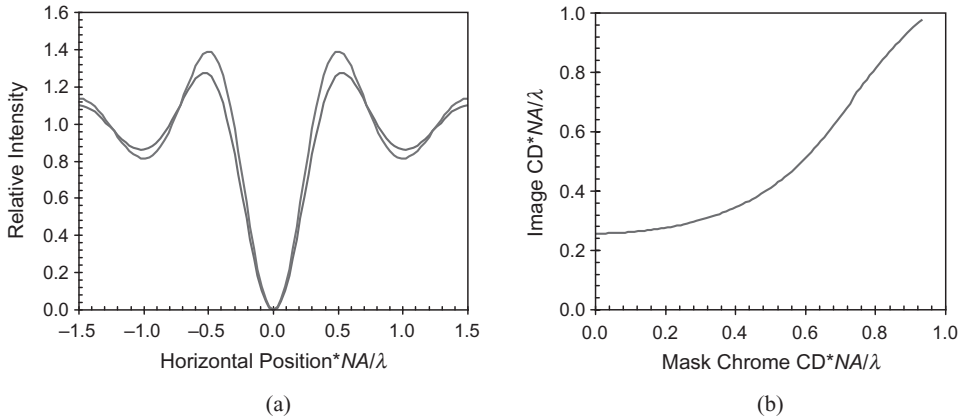


Figure 10.26 Behavior of an isolated phase-shifted line as a function of the chrome line width: (a) coherent aerial images for $w = 0$ and 50 nm , and (b) the aerial image width (at an intensity threshold of 0.25) as a function of the mask chrome width

a constant value of about $0.26\lambda/NA$ (see section 10.5.3). Setting $w = 0$ in Equation (10.11), we obtain the coherent TE aerial image for this isolated phase edge:⁹

$$I(x) = \frac{4}{\pi^2} \text{Si}^2\left(\frac{2\pi NAx}{\lambda}\right) \quad (10.12)$$

As Figure 10.26 and Equation (10.12) show, a 0 – 180° phase edge will print as a narrow dark line (in fact, the class of *strong* phase-shifting masks can be defined as those masks that use a phase edge to print a dark line). While this can be very good if a narrow dark line is desired, it can be very bad if the phase edge occurs incidentally and a dark line is not desired at that point. As will be discussed in the following section, it can be very difficult to ensure that no unwanted phase edges appear in a phase-shifted layout.

10.4.2 Phase Conflicts

‘Strong’ phase-shift masks, such as alternating PSM, have seen only limited use in manufacturing, despite their potential for nearly doubling resolution and extending DOF by even more. Strong shifters invariably use etched quartz to create the phase shift and require two mask writing steps (including an alignment in between). While the much greater complexity and expense of making alternating PSMs have certainly been an impediment to their adoption, it is not the mask manufacturing that is the biggest problem, but the mask design. Alternating PSM works by shifting the phase of the clear region to one side of a small line by 180° relative to the phase of the light coming from the other side of the line. While simple in concept, attempting to phase-shift an arbitrary layout of lines will invariably lead to *phase conflicts*.

There are two basic types of phase conflicts, as seen in Figure 10.27: no phase shift where you want it, and a phase shift where you don’t want it. The first type (Figure 10.27a) results in a lack of phase shift across a critical feature when there is an odd wrapping of phase assignments. This ‘nonshifted’ feature will not properly print. The second type (Figure 10.27b) is also called the termination problem since it usually occurs at the end of a line. Alternating phase across each side of a line will result in those two phases

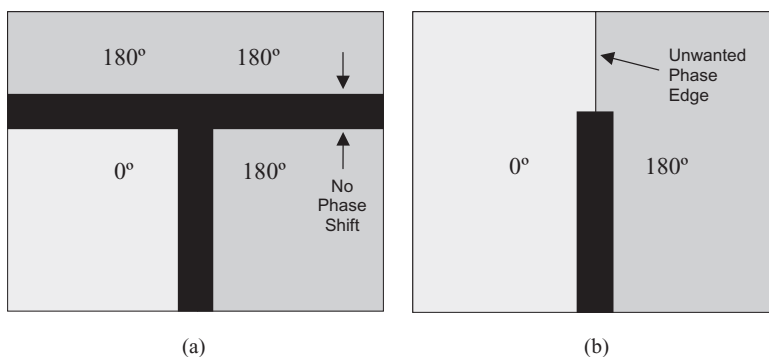


Figure 10.27 Types of phase conflicts: (a) no phase shift across a critical pattern, and (b) the phase termination problem producing an unwanted phase edge

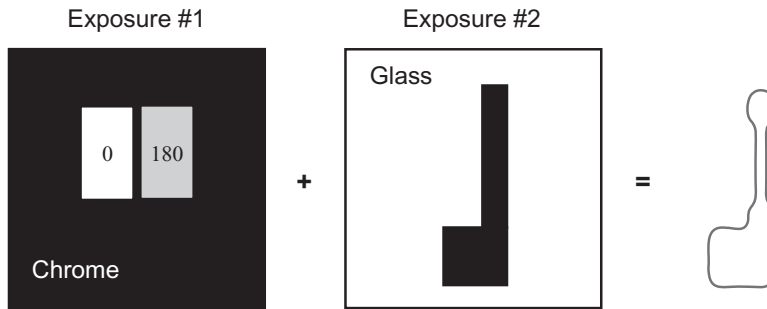


Figure 10.28 Simple example of a double-exposure alternating phase-shift mask approach to gate-level patterning

meeting at the line end. Whenever two opposing phases meet, a dark interference line is created causing a resist line to print. (In fact, as seen in the previous section, when printing small lines, the use of chrome is almost superfluous – it is the 0 to 180° phase transition that causes the dark lines to print.)

The phase conflict problems do not have an obvious solution. Taking an arbitrary layout and coloring the clear regions with two phase colors will lead to numerous conflicts in all but a few special cases. One solution, though not yet proven, is to force the original layout to be ‘phase friendly’, a layout where no phase conflicts can occur. It is unclear whether such a solution can in fact be developed and if so, what the design trade-offs will be. The alternative is costly but more practical: use two exposures from two masks. The phase-shift mask is used to define the critical features using a dark-field background to avoid phase conflicts. A second exposure uses a second bright-field mask that prints the noncritical features as well as the open areas not exposed with the first dark-field mask (Figure 10.28). Besides the obvious cost disadvantage due to the lower lithography tool throughput, this double-exposure approach also tends to have less-than-optimal transistor density (small lines can be printed, but not as close together as desired), and has not been proven to be practical on all critical levels. The double-exposure PSM process has been used successfully for polysilicon gate layers on microprocessors, where gate CD control has a profound effect on the value of the device.

10.4.3 Phase and Intensity Imbalance

The most obvious way of shifting the phase of light on a photomask is to change the thickness of quartz that one ray of light must pass through compared to another ray. This is most easily done by etching the quartz under one space by a set depth d while leaving the quartz of an adjacent space unetched (Figure 10.29). Equation (10.5) tells us the amount of phase shift as a function of the etch depth of this trench. This equation can also tell us how an error in etch depth turns into an error in phase. For 193-nm lithography, that translates into about 0.9° phase error per nanometer etch depth error.

These previous equations relating phase change to etch depth make an important assumption – that the light is traveling vertically. However, as light goes through the etched quartz hole, it begins to diffract at the bottom of the hole and its directions deviate

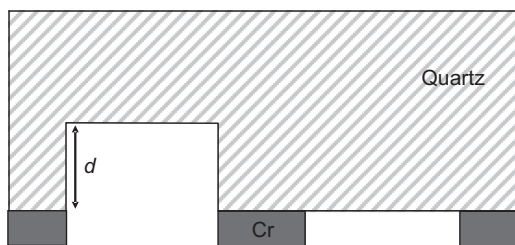


Figure 10.29 Example of a simple alternating phase-shift mask manufacturing approach

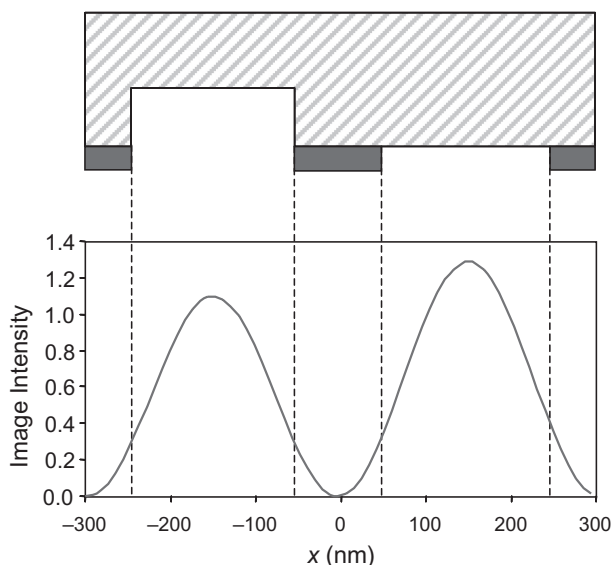


Figure 10.30 Intensity imbalance shown for an alternating phase-shift mask of equal lines and spaces

from a straight line. The smaller the hole, the greater this diffraction effect. This causes a difference in the actual phase of the light compared to the geometric ‘straight line’ approximation, and this difference is a function of the size of the etched space. Diffraction through the etched space causes a second problem: some of the diffracted light doesn’t make it out of the hole. As a result, the etched quartz space appears dimmer than the unetched space, creating an intensity imbalance in addition to the phase error (Figure 10.30). As with the phase error, the degree of intensity imbalance is spacewidth dependent.

The real, physical effects of trying to create a 180° phase shift by etching the quartz as described above are detrimental to lithography. The intensity imbalance causes the two adjacent spaces to print as different linewidths, and the phase imbalances cause these spaces to have different best-focus settings. There are a number of possible ways to fix

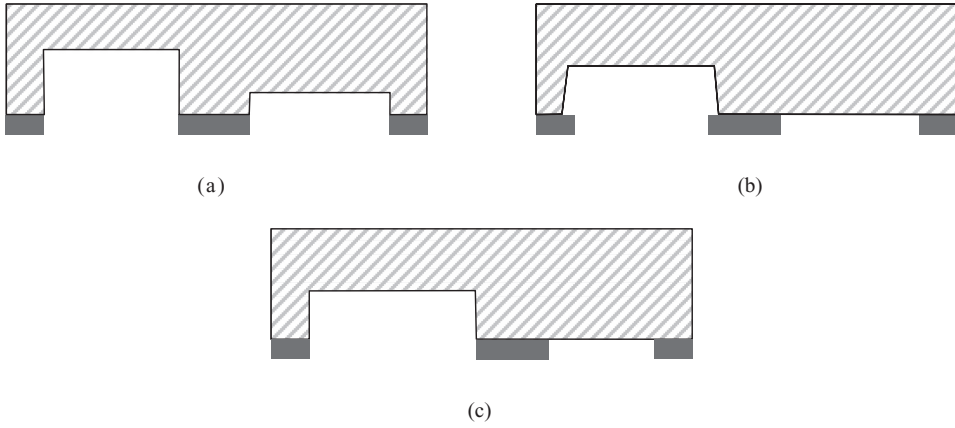


Figure 10.31 Different approaches for fixing the phase error and intensity imbalance in alternating PSM: (a) dual trench, (b) undercut etch and (c) biased space

these problems. In the *dual trench* method (Figure 10.31a), both the unshifted and shifted spaces are first etched to some depth, then the shifted space is further etched to create the desired phase shift. This approach can reduce both the phase errors and the intensity imbalance, though not perfectly for all space widths, and it complicates the mask-making process. In the *undercut etch* method (Figure 10.31b), the shifted space is etched somewhat isotropically, causing some undercut of the chrome. This widening of the etched space reduces the intensity imbalance appreciably. However, to get the intensity imbalance to approach zero, the amount of undercut can become excessive, causing possible reliability problems with the overhanging chrome. Finally, the *biased space* approach (Figure 10.31c) uses an OPC tool to bias the etched space larger in order to eliminate the intensity imbalance. This bias can be adjusted as a function of the space size and pitch in order to eliminate intensity imbalance across the full range of features. In general, none of these approaches is perfect at eliminating the phase error as a function of size and pitch.

The most common approach to alternating PSM manufacturing is to combine the biased space approach with a small amount of undercut. While not perfect, it can provide very good intensity balance and minimum phase error across all pitches.

10.4.4 Attenuated PSM

Consider a general repeating line/space transmittance function (with pitch p) where the first feature (width w_1) has amplitude and phase transmittance of t_1 and ϕ_1 , respectively, and the second feature (width $w_2 = p - w_1$) has amplitude and phase transmittance of t_2 and ϕ_2 (see Figure 10.32). The resulting diffraction pattern will be

$$T_m(f_x) = \sum_{j=-\infty}^{\infty} a_j \delta\left(f_x - \frac{j}{p}\right) \quad (10.13)$$

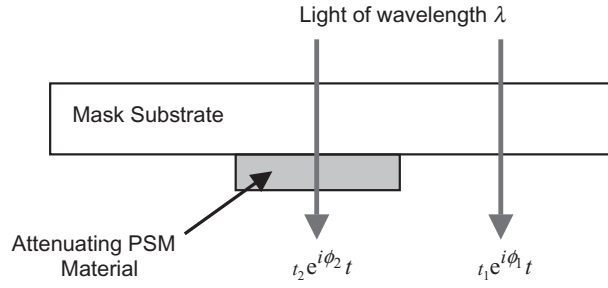


Figure 10.32 Cross section of an attenuated PSM showing how the transmitted amplitude and phase of the light is modified by the attenuating material

where

$$a_j = t_1 e^{i\phi_1} \left(\frac{\sin(j\pi w_1/p)}{j\pi} \right) + t_2 e^{i\phi_2} (-1)^j \left(\frac{\sin(j\pi w_2/p)}{j\pi} \right) \quad (10.14)$$

or,

$$\begin{aligned} a_0 &= (t_1 e^{i\phi_1} - t_2 e^{i\phi_2}) \left(\frac{w_1}{p} \right) + t_2 e^{i\phi_2} \\ a_j &= (t_1 e^{i\phi_1} - t_2 e^{i\phi_2}) \left(\frac{\sin(j\pi w_1/p)}{j\pi} \right), \quad j \neq 0 \end{aligned} \quad (10.15)$$

Without loss of generality, it will be convenient to let $\phi_1 = 0$ so that ϕ_2 represents the phase difference between the two features. If $t_1 = 1$ and $t_2 = 0$, we have the conventional chrome-on-glass result. If we let $t_1 = 1$ but allow $t_2 > 0$, the phase of this second feature will now matter. Given our experience above, we will let this phase shift be 180° . Thus, the diffraction orders are

$$\begin{aligned} a_0 &= (1 + t_2) \left(\frac{w_1}{p} \right) - t_2 \\ a_j &= a_{-j} = (1 + t_2) \left(\frac{\sin(j\pi w_1/p)}{j\pi} \right), \quad j \neq 0 \end{aligned} \quad (10.16)$$

A mask of this type is called an *attenuated PSM* (also known as an embedded PSM, or EPSM, and sometimes called a half-tone PSM).

Equation (10.16) shows the impact of allowing the nominally dark feature to transmit some light that is phase-shifted compared to the bright feature: the zero order is decreased, the first order is increased. Consider the simple case of coherent three-beam imaging where the aerial image (for TE illumination) is

$$I(x) = a_0^2 + 2a_1^2 + 4a_0a_1 \cos(2\pi x/p) + 2a_1^2 \cos(4\pi x/p) \quad (10.17)$$

For the case of equal lines and spaces, the NILS at the nominal line edge is

$$NILS = 4\pi \frac{a_1}{a_0} \quad (10.18)$$

For an attenuated PSM, using Equation (10.16) evaluated for equal lines and spaces, the NILS becomes

$$NILS = 8 \left(\frac{1+t_2}{1-t_2} \right) \quad (10.19)$$

As the transmittance of the attenuated line increases, the NILS increases as well.

While the discussion above shows the advantages of an attenuated PSM for three-beam imaging, a high-resolution lithography process will often use off-axis illumination in order to improve depth of focus for small pitch patterns. For this case, a coherent two-beam image (for TE illumination) is

$$I(x) = a_0^2 + a_1^2 + 2a_0a_1 \cos(2\pi x/p) \quad (10.20)$$

For the case of equal lines and spaces, the NILS at the nominal line edge is

$$NILS = 2\pi \frac{a_0a_1}{a_0^2 + a_1^2} \quad (10.21)$$

For the attenuated PSM case,

$$NILS = \frac{(1-t_2)(1+t_2)}{\frac{(1-t_2)^2}{4} + \frac{(1+t_2)^2}{\pi^2}} \quad (10.22)$$

For the three-beam imaging case, the best NILS is obtained by making the 0th order small and the 1st order big. For the two-beam case, maximum NILS comes from making the two orders equal in amplitude. To achieve this maximum,

$$\frac{(1-t_2)}{2} = \frac{(1+t_2)}{\pi} \quad \text{or} \quad t_2 = \frac{\pi-2}{\pi+2} \approx 0.222 \quad (10.23)$$

The above equations relate diffraction orders, images and the NILS to the electric field transmittance of the attenuated PSM material. It is common, however, to specify an attenuated PSM by its intensity transmittance, equal to $(t_2)^2$. Thus, for the two-beam imaging equal line/space patterns, the optimum intensity transmittance of the PSM is 4.93%.

Note that the transmittance used in the above equations was set to 1.0 for the transmittance of the mask substrate. Thus, the attenuated material transmittance is defined relative to the transmittance of the mask substrate. It is common, however, for the manufacturers of mask blanks to specify the transmittance of the EPSM blank in absolute terms. Letting t_{blank} be the transmittance of the EPSM coated substrate, and t_{glass} the transmittance of the glass or fused silica substrate without EPSM material, then

$$t_2 = \frac{t_{\text{blank}}}{t_{\text{glass}}} \quad (10.24)$$

For example, a common commercially available EPSM blank has an intensity transmittance of 6%. The fused silica substrate itself has a transmittance of approximate 92% at 248nm and about 91% at 193nm. Thus, the relative intensity transmittance

of the EPSM features is actually 6.5 and 6.6% for 248- and 193-nm wavelengths, respectively.

Attenuated PSM has been very widely adopted for contact and via printing, and is fairly mainstream for other critical lithography layers as well. This type of PSM is generally called a ‘weak’ shifter – it uses phase-shifting to improve the image at the edge, but does not define the edge by a 0–180° phase transition. Its great advantage is the simplicity and low cost of replacing chrome-on-glass (COG) masks, the nonphase-shift alternative. Essentially, an existing design based on COG can be converted to an EPSM by simply recalibrating the optical proximity correction models used to apply OPC to the design. Mask manufacturing, while certainly more difficult than COG, is not dramatically different (the chrome is replaced by a more complex absorber such as molybdenum silicon) and only somewhat more costly. Attenuated PSM does not suffer from phase conflicts and exhibits much reduced phase and intensity variation as a function of feature size as compared to alternating PSM. When coupled with off-axis illumination and SRAFs, it can provide most of the benefit of the much more difficult and expensive alternating PSM alternative.

One problem with the use of attenuated PSM is the occurrence of sidelobes. Figure 10.33 shows an image of an isolated space ($k_1 = 0.48$) using a 6% attenuated PSM. Interference between light from the space and light from the background EPSM material cancels out near the space edge to make the intensity go to zero and sharpen the image near the edge (i.e. increase the NILS). However, away from the edge diffracted light from the unshifted space can add constructively with the EPSM transmission to give a bright region called a *sidelobe*. Even though the intensity transmittance in this attenuated area is only 6%, the brightness of this sidelobe is 17% (in this case). A higher level of EPSM transmittance is likely to cause the sidelobe to print, resulting in catastrophic failure. Sidelobes can be suppressed, however, using strategically placed assist slots. Since the phase of the sidelobe electric field is about 180° with respect to the unshifted transmittance of a space, placing an unshifted assist space (small enough in size so that it won’t print) at the location of the sidelobe will have the paradoxical effect of reducing the intensity at that point.

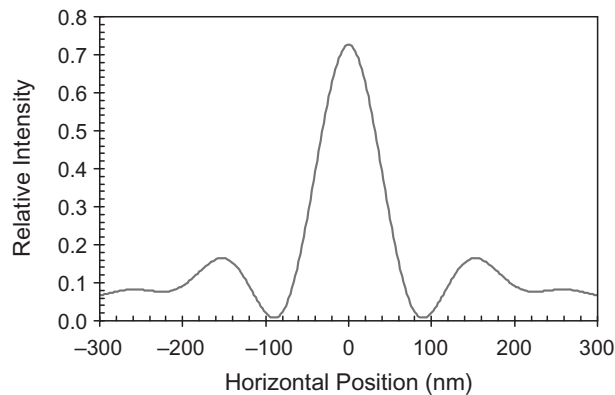


Figure 10.33 An isolated space (100nm, $NA = 0.93$, $\lambda = 193\text{nm}$, $\sigma = 0.5$) imaged from a 6% attenuated PSM mask showing sidelobes

10.4.5 Impact of Phase Errors

All of the analyses of phase-shifting masks above assume that the mask was designed and manufactured to give the ideal 180° phase shift. However, in practice there will invariably be phase errors. It will be important to understand the impact of phase errors on lithographic performance and the magnitude of these errors that can be tolerated. To investigate the impact of phase errors, the three analytical cases described in the previous sections (an isolated phase edge, an alternating PSM of lines and spaces, and an attenuated PSM pattern of lines and spaces) will be analyzed.

Consider an arbitrary isolated edge pattern oriented in the y -direction that has left and right sides with amplitude and phase transmittances of t_1 , ϕ_1 and t_2 , ϕ_2 , respectively. Looking first at the simple case of coherent illumination, the resulting aerial image electric field will be

$$E_{\text{edge}}(x) = \frac{1}{2}(t_1 e^{i\phi_1} + t_2 e^{i\phi_2}) + \frac{1}{\pi}(t_1 e^{i\phi_1} - t_2 e^{i\phi_2}) \text{Si}\left(\frac{2\pi N A x}{\lambda}\right) \quad (10.25)$$

where $x = 0$ is the position of the edge. For an ideal phase edge, $t_1 = t_2 = 1$, $\phi_1 = 0$, and $\phi_2 = \pi$ (180°). For a phase edge with phase error, we can let $\phi_2 = \pi + \Delta\phi$. Using these values in Equation (10.25) and squaring the magnitude of the electric field to obtain the intensity of the aerial image,

$$I_{\text{phase-edge}}(x) = \frac{1}{2}(1 - \cos \Delta\phi) + (1 + \cos \Delta\phi) \frac{2}{\pi^2} \text{Si}^2\left(\frac{2\pi N A x}{\lambda}\right) \quad (10.26)$$

For small phase errors, the small angle approximation to the cosine can be used,

$$\cos \Delta\phi \approx 1 - \frac{\Delta\phi^2}{2} \quad (10.27)$$

giving

$$I_{\text{phase-edge}}(x) = \frac{\Delta\phi^2}{4} + \left(1 - \frac{\Delta\phi^2}{4}\right) \frac{4}{\pi^2} \text{Si}^2\left(\frac{2\pi N A x}{\lambda}\right) \quad (10.28)$$

Looking closely at Equation (10.28), it is interesting to observe that the effect of the phase error is to take the ideal image (when $\Delta\phi = 0$) and add a uniform (d.c.) flare (see Chapter 3) in the amount of $\Delta\phi^2/4$. Like flare, a phase error for this feature will add a background dose, in this case caused by the incomplete cancellation of the out-of-phase light across the edge.

To put some numerical relevance to this relationship, an 11.5° phase error is equivalent to a 1% flare level. Since flare levels of 1% or more are commonly tolerated in lithographic projection tools, one would expect phase errors of 11.5° or more should be easily tolerated. While the above analysis assumes coherent illumination, simulation of partially coherent imaging of the same mask feature has shown the same relationship between phase error and flare (see Figure 10.34).

It is well known that phase errors also cause a change in the response of a phase-mask feature to focus errors. Again using simulation, the effect of phase errors on the process

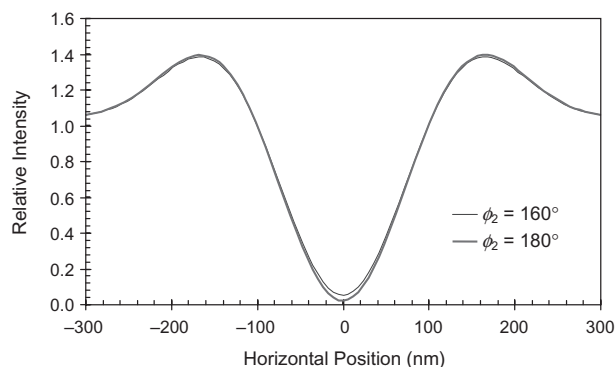


Figure 10.34 Impact of a 20° phase error on the aerial image of an isolated phase edge ($\lambda = 248\text{ nm}$, $NA = 0.75$, $\sigma = 0.5$)

window of an isolated phase edge is no different than the effect of flare on the process window. However, a new effect is introduced. When the phase error is not zero, going out of focus produces an asymmetry in the image that results in an effective image placement error. This image placement error varies linearly with defocus, acting just like a telecentricity error. Over the full range of focus within the process window, the impact on image placement error is about $\pm 5\text{ nm}$ for a 10° phase error.

Consider now an alternating PSM pattern of lines and spaces with spacewidth w_s and linewidth w_l . Assuming that the lines have zero transmittance, we shall let two adjacent spaces have arbitrary transmittances and phases of t_1, ϕ_1 and t_2, ϕ_2 , respectively. For high-resolution patterns, only the zero and first diffracted orders will pass through the lens and be used to generate the aerial image. Defining a coordinate system with $x = 0$ at the center of the first space and letting $p = w_s + w_l$ be the pitch, the amplitude of the zero and first diffraction orders will be given by

$$\begin{aligned} a_0 &= \frac{w_s}{2p} (t_1 e^{i\phi_1} + t_2 e^{i\phi_2}) \\ a_1 = a_{-1} &= \frac{\sin(\pi w_s/2p)}{\pi} (t_1 e^{i\phi_1} - t_2 e^{i\phi_2}) \end{aligned} \quad (10.29)$$

The electric field of the aerial image (assuming coherent illumination for simplicity) will be

$$E(x) = a_0 + 2a_1 \cos(\pi x/p) \quad (10.30)$$

For an ideal alternating PSM, $t_1 = t_2 = 1$, $\phi_1 = 0$, and $\phi_2 = \pi$ (180°). For a mask with phase error, we can let $\phi_2 = \pi + \Delta\phi$. Using these values in Equation (10.29),

$$\begin{aligned} a_0 &= \frac{w_s}{2p} (1 - e^{i\Delta\phi}) \approx -i\Delta\phi \left(\frac{w_s}{2p} \right) \\ a_1 &= \frac{\sin(\pi w_s/2p)}{\pi} (1 + e^{i\Delta\phi}) \approx \frac{2 \sin(\pi w_s/2p)}{\pi} \left(1 + \frac{i\Delta\phi}{2} \right) \end{aligned} \quad (10.31)$$

where the approximate relations on the right make use of a small angle approximation. Using these values in Equation (10.30) and squaring the magnitude of the electric field to obtain the intensity of the aerial image,

$$I_{\text{Alt-PSM}}(x) \approx \left(1 - \frac{\Delta\phi^2}{4}\right) I_{\text{ideal}}(x) + \Delta\phi^2 \left(\frac{w_s}{2p}\right)^2 \quad (10.32)$$

where I_{ideal} is the image of the alternating PSM for no phase error. Just as in the case of the phase edge discussed above, the impact of phase error on the image of an alternating PSM looks just like the addition of flare to the image. In this case, however, the amount of effective flare is reduced by the square of the shifted space duty ratio of the mask ($w_s/2p$). For equal lines and spaces, $w_s/2p = 1/4$ so that the effective flare for this Alt-PSM pattern is four times smaller than the effective flare for an isolated phase edge.

It would seem, then, that the wonderful properties of the alternating phase-shifting mask make them relatively immune to the small phase errors that will inevitably occur during mask fabrication. Unfortunately, a closer look reveals a more insidious problem. When the image goes out of focus, any phase error in the mask will interact with the defocus to cause an asymmetry between the shifted and unshifted spaces. The culprit lies in the zero order.

For no phase error, the zero order is completely missing ($a_0 = 0$). As a result, the image is formed completely by the interference of the two first orders. Since these orders are evenly spaced about the center of the lens, they have a natural immunity to focus errors, making improved depth of focus one of the most attractive qualities of the alternating phase mask. As the zero order grows (with increased phase error on the mask), this third beam ruins the natural symmetry of the first orders and adds a focus dependency. Again for the case of coherent illumination and assuming small phase errors, the aerial image when out of focus will be

$$I_{\text{out-of-focus}}(x, \delta) \approx I_{\text{in-focus}}(x) + 2\Delta\phi \left(\frac{w_s}{2p}\right) E_{\text{ideal}}(x) \sin(\pi\delta\lambda/p^2) \quad (10.33)$$

where E_{ideal} is the electric field for the no phase error, no defocus case [i.e. Equation (10.30) with $\Delta\phi = 0$].

Obviously, the interaction of phase error $\Delta\phi$ and defocus distance δ adds an error term to the in-focus aerial image as shown in Equation (10.33). The nature of this error term lies in the nature of the electric field E_{ideal} . Unlike the intensity, the electric field can be negative. In fact, the electric field will be negative under the phase-shifted spaces and will be positive under the unshifted spaces. Thus, for a positive focus error ($\delta > 0$) and phase error ($\Delta\phi > 0$), the error term in Equation (10.33) will make the shifted space dimmer and the unshifted space brighter. For a negative focus error, the opposite will be true. This effect is largest for smallest pitches p as can be seen from the argument of the sine term. This asymmetry between adjacent spaces through focus is the most detrimental effect of phase errors for an alternating phase-shifted mask and is illustrated in Figure 10.35.

The lithographic consequences of this intensity imbalance of adjacent spaces through focus are quite interesting. From the perspective of the space, the brighter space will print

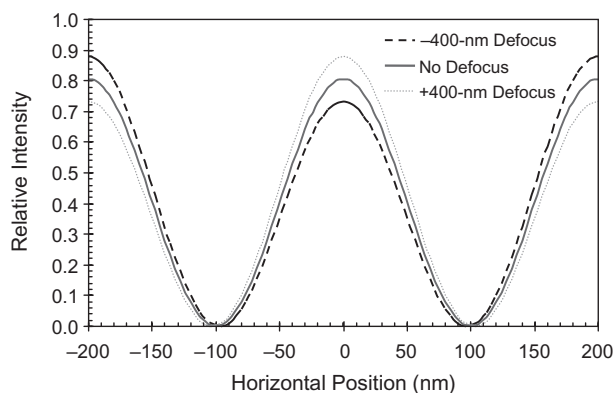


Figure 10.35 Aerial images for an alternating phase-shifting mask with a 10° phase error for +400-nm defocus (dots), no defocus (solid), and -400-nm of defocus (dashed). (100-nm lines and spaces with $\lambda = 248$ nm, coherent illumination, $0.25 < k_1 < 0.5$)

wider (assuming a positive resist), so that one space will be wider on one side of focus while its neighbor will be wider on the other side of focus. Things look quite different from the perspective of the line, however. As the space to one side of the line gets wider with a focus error, the space to the other side of the line will get narrower. As a result, the linewidth remains about constant but its position shifts away from the brighter space and toward the dimmer space. Thus, the impact of the interaction of phase and defocus errors for an alternating PSM can be seen as an error in the placement of the line in the PSM array. When both transmission and phase errors are present, the two effects on space image brightness can cancel at some defocus. The resist image with the best fidelity would then appear out of focus.

While alternating aperture PSMs are important, the overwhelming majority of phase-shift masks used today are attenuated PSMs. How does a small phase error affect the lithographic performance of an attenuated PSM? Consider an equal line/space pattern where the line has an electric field amplitude and phase transmittance of t and ϕ , respectively. For a mask with phase error, we can let $\phi = \pi + \Delta\phi$. Using this value in Equation (10.15), and assuming that the phase error is small,

$$\begin{aligned}
 a_0 &= \frac{1}{2}(1 - te^{i\Delta\phi}) = \frac{1}{2}(1 - t \cos(\Delta\phi)) - i\frac{1}{2}t \sin(\Delta\phi) \approx \frac{1}{2}\left(1 - t + \frac{t\Delta\phi^2}{2}\right) - i\frac{1}{2}t\Delta\phi \\
 a_1 &= a_{-1} = \frac{1}{\pi}(1 + te^{i\Delta\phi}) \approx \frac{1}{\pi}\left(1 + t - \frac{t\Delta\phi^2}{2}\right) + i\frac{1}{\pi}t\Delta\phi
 \end{aligned} \tag{10.34}$$

Calculating the magnitude and the phase of each diffraction order,

$$\begin{aligned}
 |a_0| &\approx |a_0|_{\text{ideal}} + \frac{\Delta\phi^2}{4}\left(\frac{t}{1-t}\right) \\
 \angle a_0 &\approx -\Delta\phi\left(\frac{t}{1-t}\right)
 \end{aligned} \tag{10.35}$$

$$\begin{aligned}
 |a_1| &\approx |a_1|_{\text{ideal}} - \frac{\Delta\phi^2}{2\pi} \left(\frac{t}{1+t} \right) \\
 \angle a_1 &\approx \Delta\phi \left(\frac{t}{1+t} \right)
 \end{aligned}
 \tag{10.36}$$

Let's investigate the impact of the changes in the magnitude and phase of each diffracted order separately. As shown in Equations (10.35) and (10.36), the magnitudes of the orders vary as the phase error squared (and so should be quite small for small errors). As an example, for a 6% EPSM with a 10° phase error, the magnitudes of the zero and first orders change by only +0.7 and -0.2%, respectively. The resulting impact on the aerial image is quite small, less than 0.2% intensity difference in most cases.

The phase of the diffraction orders, on the other hand, varies directly as the EPSM phase error. In fact, the phase difference between the zero and first orders, which ideally would be zero, becomes in the presence of EPSM phase error

$$\angle a_1 - \angle a_0 \approx \frac{2\Delta\phi t}{1+t^2} \tag{10.37}$$

What is the impact of such a change in the phase difference between the diffraction orders? Focus also causes a phase difference between the zero and first orders. For the simple case of coherent illumination (that is, three-beam imaging), a defocus of δ causes a phase difference of

$$\angle a_1 - \angle a_0 \approx \frac{\pi\delta\lambda}{p^2} \tag{10.38}$$

Thus, the effect of the EPSM phase error will be to shift best focus by an amount given by

$$\delta_{\text{shift}} \approx \frac{2p^2\Delta\phi t}{\pi\lambda(1+t^2)} \tag{10.39}$$

Possibly a more useful expression for this effective focal shift is as a function of the depth of focus. Using the simple Rayleigh criterion for DOF,

$$DOF \approx k_2 \frac{p^2}{\lambda} \tag{10.40}$$

where k_2 is some number less than 1 (0.7 is reasonable). Combining these expressions, and assuming that $t^2 \ll 1$,

$$\delta_{\text{shift}} \approx \frac{2}{\pi k_2} \Delta\phi t DOF \approx \Delta\phi t DOF \tag{10.41}$$

For a 6% EPSM, focus will shift by about 0.4% of the DOF per degree of EPSM phase error. Remembering that Equation (10.39) was derived under the simple assumption of coherent illumination, full image simulations show that the use of partial coherence can double or triple the focus shift compared to the coherent case. Thus, a 10° phase error

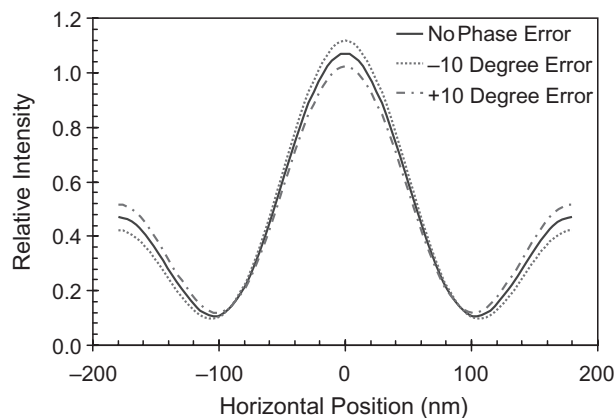


Figure 10.36 A small phase error in an EPSM mask changes the aerial image in the same way as a small shift in focus. Here, $\pm 10^\circ$ phase error moves the image closer and farther away from best focus (wavelength = 248 nm, NA = 0.8, 180-nm lines/space pattern, coherent illumination, 150-nm defocus). For this case, a 10° phase error shifts best focus by about 14 nm

might cause best focus to shift by 10% of the DOF, potentially causing a 20% loss in DOF. Off-axis illumination, however, tends to lower this effect since this illumination is specifically intended to minimize the impact of phase errors between the zero and first orders. Figure 10.36 illustrates this focus-shift effect for three-beam imaging, showing also that unlike alternating PSM, there is no pattern placement change through focus in the presence of an EPSM phase error.

In general, attenuated phase-shift masks are much less sensitive to phase errors than alternating phase-shift masks. The major impact of small phase errors is a focus shift, so the biggest worry would be a variation of EPSM phase across a reticle rather than a mean to target error. Since cleaning can cause phase errors in an EPSM mask, understanding the impact of these errors is quite important.

10.5 Natural Resolutions

The discussion above defined feature resolution with respect to CD control, (using depth-of-focus as the metric of CD control) and pitch resolution with respect to the frequency cutoff of the imaging lens. There are, however, three special purpose metrics of resolution that can also be used to define the ultimate capabilities of an imaging system. These special metrics, though defined for very specific mask features, have almost universal appeal as ‘natural’ metrics of imaging resolution.

10.5.1 Contact Holes and the Point Spread Function

The first metric concerns the smallest possible contact hole that can be printed. Consider a mask pattern of an isolated contact hole in a chrome (totally dark) background. Now

let that hole shrink to an infinitesimal pinhole. If this small pinhole could be imaged in a positive resist, how big would the resist hole be? Of course, there is a practical problem with experimentally determining this ‘resolution’: as the pinhole shrinks, the intensity of light reaching the wafer becomes infinitesimally small, making the required exposure time for the resist grow to infinity. However, from a theoretical perspective, we can avoid this problem by assuming we are very patient and calculating the printed result.

Thinking first of just the imaging tool, what would be the aerial image resulting from this infinitely small pinhole? We will normalize our aerial image coming from the pinhole to have a peak intensity of 1.0 when in-focus for an ideal, aberration-free optical system. The aerial image of a pinhole, when normalized in this way, is called the *point spread function* (PSF) of the optical system (see Chapter 2). The PSF is a widely used metric of imaging quality for optical system design and manufacture, and is commonly calculated for lens designs and measured on fabricated lenses using special benchtop equipment. For classical imaging applications, the PSF can be calculated as the square of the magnitude of the Fourier transform of the imaging tool exit pupil. For an in-focus, aberration-free system of wavelength λ , the pupil function is just a circle whose radius is given by the numerical aperture NA , and the PSF becomes

$$PSF_{\text{ideal}} = \left| \frac{J_1(2\pi\rho)}{\pi\rho} \right|^2 \quad (10.42)$$

where J_1 is the Bessel function of the first kind, order one, and ρ is the radial distance from the center of the image normalized by multiplying by NA/λ .

How wide is the PSF? For large contact holes, the normalized intensity at a position corresponding to the mask edge (that is, at the desired contact hole width) is about 0.25–0.3. If we use this intensity range to measure the width of the PSF, the result is a contact hole between 0.66 and $0.70\lambda/NA$ wide. Thus, this width represents the smallest possible contact hole that could be imaged with a conventional chrome-on-glass (i.e. not phase-shifted) mask. If the contact hole size on the mask approaches or is made smaller than this value, the printed image is controlled by the PSF, not by the dimensions of the mask. Making the contact size on the mask smaller only reduces the intensity of the image peak. Thus, this width of the PSF, the ultimate resolution of a chrome-on-glass contact hole, is a natural resolution of the imaging system.

If instead of using a chrome-on-glass mask when making our conceptual pinhole an attenuated PSM is used, the width of the resulting PSM is decreased. Assuming that the mask background electric field transmittance is t and the phase shift is exactly 180° , the resulting PSF is

$$PSF_{\text{ideal}} = \left| \frac{\frac{J_1(2\pi\rho)}{\pi\rho} - t}{1 - t} \right|^2 \quad (10.43)$$

A graph of this PSF for $t = 0.245$ (a 6% intensity transmittance ESPM) compared to $t = 0$ (a chrome-on-glass mask) is shown in Figure 10.37. At an intensity threshold of

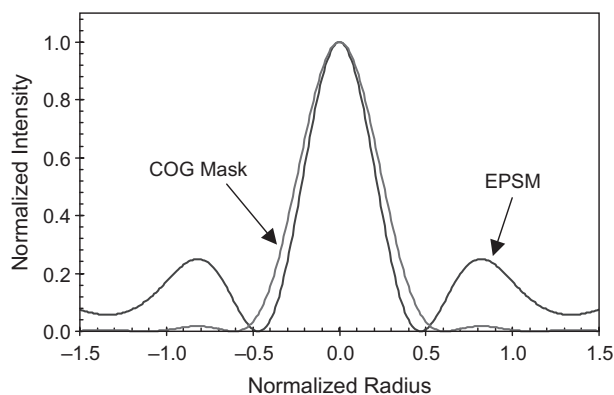


Figure 10.37 Comparison of the ideal PSF from a chrome-on-glass (COG) mask and a 6% intensity transmittance embedded phase-shifting mask (EPSM)

0.25, the COG PSF has a width of about $0.705\lambda/NA$, whereas the EPSM mask produces a PSF with a width of $0.595\lambda/NA$ (16% narrower). Note that the narrowing of the PSF comes at the expense of increased sidelobes, which reach an intensity of nearly 0.25 for this 6% EPSM mask. Thus, 6% intensity transmittance represents about the maximum practical transmittance level when printing isolated contact holes unless some sort of sidelobe suppression (in the form of strategically placed assist slots) is employed.

10.5.2 The Coherent Line Spread Function (LSF)

If instead of an infinitely small pinhole in a dark background an infinitely narrow space is used, the resulting image is called the *line spread function* (LSF). Unlike the PSF, however, the LSF image will depend on the illumination used. For coherent illumination, the LSF can be calculated as an integral over one dimension of the coherent PSF:

$$LSF(x) = \int_{-\infty}^{\infty} \frac{J_1\left(\frac{2\pi NA}{\lambda} \sqrt{x^2 + y^2}\right)}{\frac{\pi NA}{\lambda} \sqrt{x^2 + y^2}} dy \quad (10.44)$$

For incoherent illumination,

$$LSF(x) = \int_{-\infty}^{\infty} \left| \frac{J_1\left(\frac{2\pi NA}{\lambda} \sqrt{x^2 + y^2}\right)}{\frac{\pi NA}{\lambda} \sqrt{x^2 + y^2}} \right|^2 dy \quad (10.45)$$

Alternately, the coherent LSF can be calculated using the image of an isolated space. From Chapter 2, the TE coherent image of an isolated space of width w is

$$I(x) = \frac{1}{\pi^2} \left[\text{Si}\left(\frac{2\pi NA}{\lambda}(x + w/2)\right) - \text{Si}\left(\frac{2\pi NA}{\lambda}(x - w/2)\right) \right]^2 \quad (10.46)$$

Normalizing this image so that the peak intensity is 1,

$$\frac{I(x)}{I(0)} = \left[\frac{\text{Si}\left(\frac{2\pi NA}{\lambda}(x+w/2)\right) - \text{Si}\left(\frac{2\pi NA}{\lambda}(x-w/2)\right)}{2\text{Si}\left(\frac{\pi NA}{\lambda}w\right)} \right]^2 \quad (10.47)$$

Taking the limit of Equation (10.47) as w goes to zero (by applying L'Hopital's rule and realizing that the derivative of the sine integral is just the sinc function),

$$LSF(x) = \lim_{w \rightarrow 0} \frac{I(x)}{I(0)} = \left[\frac{\sin(2\pi NAx/\lambda)}{2\pi NAx/\lambda} \right]^2 \quad (10.48)$$

Using an intensity threshold of 0.25, the coherent illumination LSF has a width of $0.603\lambda/NA$, and is thus 15 % narrower than the PSF. The LSF is a natural resolution limit of an isolated space.

10.5.3 The Isolated Phase Edge

Another special mask feature which exhibits a similar natural resolution behavior is the 180° phase edge, as described previously in section 10.4.1. Consider a chromeless mask with a large region shifted by 180° to produce a long, straight boundary between the 0° and the 180° regions. Since light transmitted to either side of this edge will have a 180° phase difference, the light that diffracts and interferes under the edge will cancel out, producing a dark line centered under the phase edge. As a result, this isolated 180° phase edge will print as a narrow line in a positive photoresist.

For coherent TE illumination, the in-focus image of an isolated phase edge was given in Equation (10.12). Figure 10.38 shows a graph of this equation. The width of this image

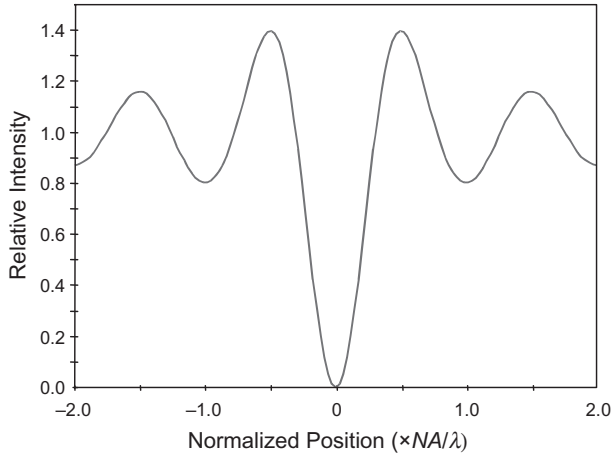


Figure 10.38 The aerial image of an isolated 180° phase edge (shown here using coherent illumination) will produce a narrow line in a positive resist

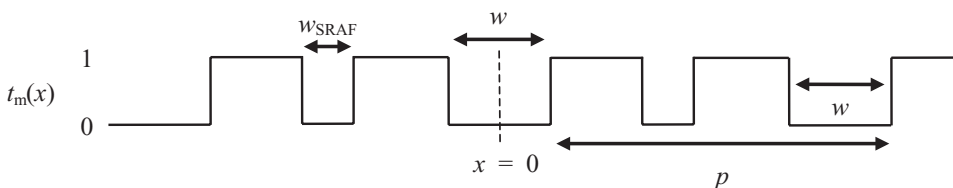
can be estimated in the same way as the PSF. Assuming an intensity level between 0.25 and 0.3, the width of the phase edge line is between 0.26 and $0.29\lambda/NA$. The use of partially coherent illumination does not appreciably change this width.

What controls the width of the image of an isolated phase edge? Obviously the edge itself does not have a ‘width’. Like the PSF and the LSF, the width of the phase edge image is in fact an inherent property of the imaging system, controlled by its numerical aperture and wavelength. In fact, analogous to the LSF, the image of Equation (10.12) could be called the *phase edge line spread function*. The size of this tiny line is the ‘natural’ resolution of any 180° phase edge. But its importance goes beyond just the printing of small isolated lines. A phase-shifted mask can be thought of as a collection of 0 – 180° phase transitions. Each transition has a tendency to print at its natural linewidth of about $0.26\lambda/NA$. Much insight can be gained by approaching phase-shifting mask design with this natural linewidth in mind.

For the special types of mask features described here, the resolution of that feature becomes a function only of the properties of the imaging system and, strangely, becomes independent of the size of the feature on the mask. I call these resolution properties the ‘natural’ resolutions of the imaging tool.

Problems

- 10.1. Figure 10.1 shows an example of how the depth of focus varies with feature size. Generate an equivalent plot using the Rayleigh DOF criterion. Assume equal lines and spaces, coherent three-beam imaging, and $k_2 = 0.8$.
- 10.2. Derive the diffraction pattern of a semi-dense line (width w , pitch p) with single SRAFs (width w_{SRAF}) placed in the center of the spaces:

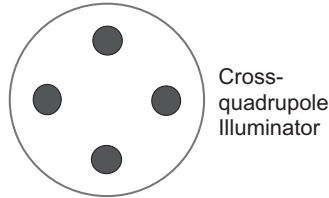


- 10.3. Considering the FEM plots of Figure 10.14 to be typical of dense and isolated lines, how does the iso-dense bias vary with focus?
- 10.4. What is the smallest value of k_{pitch} possible under the following scenarios:
 - (a) COG mask and conventional illumination as σ goes to zero (i.e. coherent illumination)
 - (b) Alt-PSM mask and conventional illumination as σ goes to zero (i.e. coherent illumination)
 - (c) COG mask and dipole illumination with very small poles
 - (d) COG mask and quadrupole illumination with very small poles
- 10.5. Explain the main advantage of dipole illumination compared to quadrupole illumination. Explain the main disadvantage of dipole illumination compared to quadrupole illumination.

- 10.6. For an optimized dipole illuminator, a two-beam aerial image results:

$$I(x) = a_0^2 + a_1^2 + 2a_0a_1 \cos(2\pi x/p)$$

Since the dipole can only print small-pitch patterns oriented in one direction, one possible solution is to add a second dipole rotated by 90° to obtain an illuminator called a *cross-quadrupole*:



When printing at the pitch for which the dipole and quadrupole are optimized, derive the resulting in-focus aerial image (assume TE illumination) for a cross-quadrupole.

- 10.7. The coherent two-beam aerial image for an alternating PSM mask is given in Equation (10.9). From this, derive the image for a chromeless repeating line/space pattern by letting the width of the chrome line in the alternating PSM mask go to zero. Qualitatively, what is the effect of reducing the chrome width?
- 10.8. Derive Equation (10.10), the diffraction pattern for an isolated phase-shifted line.
- 10.9. Derive Equation (10.11), the coherent aerial image for an isolated phase-shifted line.
- 10.10. Derive Equation (10.15) from Equation (10.14).
- 10.11. Equations (10.16) and (10.17) give the coherent (TE) image for an attenuated PSM pattern of lines and spaces. What happens to this image as t_2 goes to 1? (Such a mask is called a chromeless PSM.)
- 10.12. Derive an equation for the coherent LSF when using an attenuated PSM. What is the maximum sidelobe intensity for the LSF for a 6% attenuating PSM?
- 10.13. Derive an expression for the normalized image log-slope of the isolated phase edge, assuming the nominal feature size is at an intensity threshold of 0.25.

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